Agency Incentives and Disparate Revenue Collection: Evidence from Chicago Parking Tickets*

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Abstract

We leverage a sharp 2012 parking fine increase for failing to purchase vehicle registration to examine disparate ticketing patterns across enforcement agencies in Chicago. Using an event-study framework, we find that Chicago police increased their enforcement of car registration non-compliance in Black relative to non-Black neighborhoods, with no observed disparate response for non-police enforcement agencies. This disparity is unexplained by compliance and is instead driven by departmental revenue incentives and multitasking-driven lower marginal search costs in Black neighborhoods. Disparate enforcement also exacerbated existing gaps in financial instability, generating increased rates of ticket non-payment and bankruptcy filings in Black neighborhoods.

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1 Introduction

Parking fines are major revenue generators for United States cities. For example, in 2016, Chicago raised \$264 million from parking citations, equivalent to an annual \$97.20 per-capita tax; similarly, New York City raised \$565 million from parking fines in the year prior (Diskin, 2019; Digital Editors, 2021). The ability of a city to raise revenue from these types of fines depends on both its enforcement and collection policies and its residents' ability to pay these fines. However, reliance on fines and fees as a local revenue stream can impose significant economic consequences on residents, especially those with lower ability to pay (Makowsky and Stratmann, 2009, 2011; U.S. Department of Justice Civil Rights Division, 2015).[1]

Local governments can use many different kinds of agents to enforce parking violations, including police officers, parking enforcement aides, and other third-party contractors. The equity and efficiency of parking enforcement across neighborhoods are therefore dependent in part on the behavior of agents whom the government hires. Consequently, agents may disparately enforce violations across areas if they are maximizing a different objective function than the government - for example, minimizing search costs or focusing on alternative, non-parking enforcement responsibilities.[2] To the extent such wedges between government and agent exist and are correlated with racial and ethnic divisions within a metropolitan area, agents may respond to collection incentives in ways that generate disparate outcomes in the population while also harming revenue.

In this paper, we examine how an increase in motor vehicle registration cost, which we refer to as "stickers," and the related fine for non-compliance affected ticket enforcement patterns in Chicago. This 2012 policy increased the cost of typical vehicle registration from \$75 to \$85 (\$120 to \$135 for larger vehicles), a 13% increase, and the fine for registration non-compliance from \$120–\$200, a 67% increase. Thus, it simultaneously made compliance more expensive and enforcement more profitable relative to other parking fines. Aggressive enforcement and punishment had severe consequences for Chicago residents, particularly in predominantly Black neighborhoods, leading to claims of disparate impact.[3]

Since both Chicago Police Department (CPD) and other agents, primarily but not exclusively parking enforcement agents (henceforth, non-CPD agents), can enforce municipal

[1]Propublica estimates that unpaid parking debt alone in Chicago totals over \$1.5 billion debt (Samuelson, 2018).

[2]There is a long literature examining multi-task principal-agent models and empirically investigating agent responses in the face of differing incentives. For example, see Holmstrom and Milgrom (1991) and Aghion and Tirole (1997) for the canonical theory, and Jacob and Levitt (2003) and Knutsson and Tyrefors (2022) for empirical work in the context of teachers and ambulances, respectively.

[3]While predominantly Black neighborhoods account for only 22% of tickets, they hold 40% of all ticket debt. (Sanchez and Kambhampati, 2018).

parking laws, we separately examine the impact of the sticker tax increase on both types of agents, with the major distinction being that while non-CPD agents are evaluated on their ticketing volume, CPD officers are not. Furthermore, CPD officers have additional, broader responsibilities to "work for the benefit of its citizens by protecting life and property from harm and maintain order" (Department of Human Resources, 2023). Thus, the budget reform provides a unique opportunity to evaluate how governments use different incentives across various agents to affect revenue-generating enforcement and the downstream impact this has across residents.

We test for disparate enforcement across the two types of agents using administrative parking ticket data from 2007 to 2016 in Chicago and a difference-in-differences (DiD) framework, estimating the relative change in sticker enforcement across Black and non-Black neighborhoods.[4] Across Black versus non-Black neighborhoods, our results show consistent evidence of a disparate response for CPD-issued tickets. We find that CPD sticker enforcement increased by nearly 2,500 tickets in Black relative to non-Black neighborhoods per year. Conversely, non-CPD ticketing activity exhibits no such changes across neighborhoods.

The relative increase in sticker enforcement in Black neighborhoods also had sharply negative implications on the source of collected revenue and financial outcomes (e.g., bankruptcy) of those ticketed. At the ticket level, we find that revenue per ticket decreased by $27 in Black neighborhoods relative to non-Black neighborhoods, indicating increased instances of nonpayment and financial strain. Accordingly, we estimate that more than half of the additionally issued sticker tickets resulted in notices of nonpayment, with a further 10% resulting in bankruptcy filings.[5] Despite these low payment rates, on aggregate, the increased sticker enforcement shifted the tax burden from non-Black neighborhoods to Black neighborhoods. Relative to non-Black neighborhoods, annual collected revenue increased by over $270,000 more in Black neighborhoods, representing a regressive increase of $5 per capita. Conversely, non-CPD agents collect greater revenue from non-Black neighborhoods.

Given that both the cost of registration and the fine increased simultaneously, these disparities could reflect differential enforcement or compliance across Black and non-Black neighborhoods.[6] Using sticker purchasing data at the neighborhood level from 2008–2016, we rule out changing compliance, as we find no differential changes in purchasing behavior

[4] We also stratify across income and ability to pay and do not find similar patterns of disparate enforcement for CPD and non-CPD issued parking tickets.

[5] Over our sample period the number of Chapter 13 bankruptcies with parking debt rose substantially, from 1,000 per year to 10,000 per year. Similarly, the median parking debt per case more than doubled, from $1,500 to $3,900 (Sanchez and Kambhampati, 2018).

[6] Variation in neighborhood characteristics (e.g., number of parking meters) and its residents (e.g., number of vehicle owners with valid registration) may also lead to differential non-compliance which could warrant differential enforcement, holding fixed policing patterns.

across neighborhoods. Through a complementary decomposition exercise, we further document outsized differential enforcement in Black neighborhoods, even after accounting for differences in the number of unregistered vehicles by neighborhood. Coupled with similar purchasing responses, these results further underscore that the patterns we document are predominantly due to changes in CPD enforcement rather than simply reflecting divergence in residents' compliance with the policy.

What drives the differences in ticketing behavior across enforcement agencies? We consider two complementary potential mechanisms, finding evidence in favor of each. First, we find strong evidence that the disparate enforcement stems partly from multitasking responsibilities due to crime responses. Specifically, we show that the increase in sticker ticketing is strongest in areas with the highest crime rates, suggesting that CPD officers are increasing their ticketing effort in areas where their search costs are lowest. In contrast, non-CPD enforcement is independent of local crime rates, as such multitasking concerns are not present.

Second, we find contextual and empirical evidence that CPD faces top-down revenue collection pressures. Local officials view ticketing revenue as a practical method for revenue generation (Lentino, 2017), and memos from City Hall also raise concerns when collected ticket revenue falls below projected targets (Rivera, 2010). Empirically, we show that on the first day sticker enforcement is legally permitted, CPD officers increase their ticketing volume by nearly 50% post-reform, compared to the pre-reform level. On the other hand, non-CPD agents face performance incentives based on ticket volume, suggesting their behavior should be independent of changes to fine amounts. Accordingly, we find little evidence that non-CPD agents exhibit the same change in ticketing dynamics when sticker tickets are permissible. Moreover, in a complementary exercise leveraging the privatization of Chicago's parking meters, we show that CPD enforcement of expired meters, relative to non-CPD enforcement, falls sharply after the rights to ticketing revenue were sold to a private investment group. This response is also consistent with CPD officers altering their enforcement behavior in response to the destination of the ticketing revenue, as post-privatization, revenue accrues to the investment group rather than the city. Taken together, we view both multitasking and top-down revenue collection incentives as key drivers of the disparate and distinct enforcement patterns.

Finally, we show that disparate ticketing in response to the reform is a widespread phenomenon and not confined to a handful of "bad apple" officers. We estimate CPD officer-specific responses to the budgetary reform and show that 65-85% of officers increase their sticker ticketing activity after the fine increase. Across the officer distribution, the marginal sticker ticket is also almost twice as likely to be written in a Black neighborhood than in a non-Black neighborhood. Regressing officer characteristics against these estimated policy responses reveals few strong correlations, but the empirical patterns we observe can also not be fully explained by the neighborhood demographics of officer assignments. Instead, we conclude that the disparate enforcement across neighborhoods was part of a broader departmental phenomenon and revenue collection effort in response to the budget reform and existing deficit.

This paper builds empirical evidence on incentives and the behavior of public sector agents and their role in revenue generation for local governments.[7] We build on prior work studying the responses of police officers to pay (Mas, 2006) and opportunities for overtime compensation (Chalfin and Goncalves, 2021). We show that police officers in our setting are responsive to governmental revenue goals in ways that are not present among privately contracted agents, likely due to differences in incentives across agencies, and that this response leads to disparate revenue collection and financial outcomes in the population.[8]

In doing so, we connect to a rich literature in public finance studying how the structure of tax enforcement and laws may contribute to disparities and evasion (Allingham and Sandmo 1972; Slemrod and Yitzhaki 2002). For instance, work has shown that Black homeowners pay higher property taxes (Avenancio-León and Howard, 2022), Black taxpayers are more likely to be audited for claiming the Earned Income Tax Credit (Elzayn et al., 2025), and Black married couples are more likely to incur a "marriage penalty" in income taxes (Brown, 2022; Holtzblatt et al., 2023). We highlight an additional channel through which revenue-generating policies have disparate and regressive impacts: the enforcement of a tax on car ownership through city car registration.

Previous work has also studied the same policy variation to understand bankruptcy behavior (Morrison, Pang, and Uettwiller, 2020). Using a similar identification strategy to what we will show, Morrison, Pang, and Uettwiller (2020) document that Chicago's ticket enforcement policy change dramatically increased Chapter 13 bankruptcy filings, particularly in Black communities. They note that the drivers of Chapter 13 bankruptcy cases were not lawyer-led but driven by Chicago's overall enforcement and collection decisions. We add to this work by showing that the upstream enforcement decisions were due to sticker ticket enforcement by the CPD and not other enforcement agencies.

Finally, our study links disparate policing and the actions of government agents to fi-

[7]For theoretical work in this area, see Prendergast (2007, 2008) and for a review on incentives and decision making, see Kamenica (2012).

[8]Our results also build on a recent strand of work studying the consequences of both multitasking (Reeves 2024) and specialization (Dee and Pyne 2022; Davis et al. 2025) among police agencies. Additionally, prior work in public finance shows tax collectors are responsive to performance pay (Kahn, Silva, and Ziliak 2001; Khan, Khwaja, and Olken 2016). For more on the role of law enforcement in revenue collection, see Harvey (2020), Goldstein, Sances, and You (2020), and Makowsky, Stratmann, and Tabarrok (2019).

nancial fragility that disparately falls along racial lines.[9] In contrast to prior work which emphasizes disparate policing in highway patrol agencies (e.g., Goncalves and Mello 2021), our focus on non-moving violations and metropolitan policing implies that officer's ticketing choices are less likely to be confounded with concerns for public safety. For example, failing to stop a speeding driver in the case of highway ticketing choice could have more severe consequences than failing to ticket an improperly parked car. Thus, Chicago police are effectively being tasked as tax collectors. Furthermore, given our rich data, we can cleanly measure the monetary and revenue implications of disparate (tax) policing.[10]

2 Institutional Background and Setting

Chicago is often recognized as one of the most spatially segregated cities in the United States and has wide racial gaps in socioeconomic outcomes. For example, Black-owned businesses are valued at less than one-twelfth the value of white businesses, and the median household income for Black families ($30,303) is less than half that of white families ($70,960) (ProsperityNow, 2017). These inequalities also extend to local crime rates and rates of contact with the criminal justice system. Black neighborhoods experience substantially higher homicide rates (Sharkey and Marsteller, 2022), and Black motorists comprise the majority (60%) of traffic stops in the city (Goudie et al., 2024). Importantly for this paper, this increased exposure to law enforcement also raises the capacity for greater sticker ticket enforcement, as officers are more likely to observe any expired registrations given the higher rates of contact in Black neighborhoods. These features imply that any changes to CPD sticker enforcement efforts will also have a disproportionate impact in Black neighborhoods.

The city of Chicago relies heavily on parking ticket revenue, with 7% of its 3.6 billion dollar operating budget coming from the fines it collects. Each year, the city issues over 3 million tickets for parking violations, vehicle compliance, and automated traffic camera violations (Sanchez and Kambhampati, 2018).

Chicago's parking fines are also particularly punitive. Unpaid fines double 25 days from the initial ticket issue date and are subject to additional late fees. If a driver has repeated violations or fails to pay their fines, they can receive a vehicle seizure notice (eventually

[9] See Owens and Ba (2021) for a comprehensive literature review on policing and disproportionate burden across demographic subgroups.

[10] Recent work on financial sanctions in the justice system has found mixed evidence of the impact of these sanctions on outcomes. Pager et al. (2022) and Finlay et al. (2024) show that financial sanctions from a criminal conviction have no long-term or short-term impact on labor market or recidivism outcomes. However, Goncalves and Mello (2023), Mello (2025), Kessler (2020), and Hansen (2015) have found negative impacts on financial outcomes but reduced recidivism from harsher fines.

having their car seized) and ultimately have their license suspended.[11] Drivers can choose to enter into a payment plan with the city, but to qualify for the standard payment plan, the driver must pay a $1,000 down payment on total vehicle debt plus payment in full on any tow, boot, or storage fees. If the driver is unable to commit to a plan, they can declare bankruptcy. Chicago also has anti-scofflaw rules that prevent those with unpaid tickets or debts to the city from accessing contracts, licenses, or grants. For example, municipal jobs, such as teaching in a classroom, are inaccessible for those with unpaid parking tickets. With no statute of limitations, parking debt in Chicago can follow the driver for an entire lifetime.[12]

Column 1 of Table 1 shows the top ten parking tickets by total revenue collected from 2007 to 2011. The most issued ticket has a fine of $50. Of the top tickets, only three are related to parking (residential permit parking, expired meter, parking in prohibited areas). Four of the most popular tickets relate to having correct licensing or registration.

The most punitive parking fine is for failing to properly display the city sticker on the car windshield. The city sticker, which is colloquially known as the 'sticker tax,' is an annual registration fee that Chicago residents with vehicles must pay to own a vehicle in the city. While the registration is $75 for sedans and $120 for larger passenger vehicles, the fine for failing to buy the sticker or failing to display the sticker is almost double that of the next expensive ticket, at $120 plus a $40 late fee. In October 2011, the city announced it would be raising the registration fee for the stickers from $75 to $85 for smaller passenger vehicles (<4,500 lbs) and $120 to $135 for larger passenger vehicles (Civic Federation, 2012). Further, the fine for not paying the tax would increase by two-thirds, from $120 to $200, effective February 2012.

The fee increase was announced in conjunction with other aggressive revenue-generating policies in an attempt to close a $637 million projected deficit in the 2012 budget (Emanuel, 2011). One of these policies was an aggressive debt collection plan that directly affected how the city collected and enforced payment of parking ticket fees. This plan would allow the city to begin garnishing the wages and tax returns for high debtors in addition to suspending licenses and impounding cars. Once the maximum amount of fees had been levied, the city could garnish drivers' state tax returns and 15% of wages (Andriesen, 2012). The stated goal of this aggressive debt collection plan for parking and traffic infractions was to reduce

[11] Specifically, after three unpaid parking tickets, red light tickets, or speed camera tickets within a year or two unpaid parking tickets, red light camera tickets, or speed enforcement tickets that are one year past due, the car can be impounded or booted, and the vehicle owner receives a seizure notice. After ten or more non-moving violations or five unpaid tickets from automated red-light or speed cameras, the city of Chicago suspends the vehicle owner's driver's license.

[12] In contrast, Los Angeles and New York City have statutes of limitations that are 5 and 8 years, respectively.

employee indebtedness and to hold rental car companies accountable for their parking fines. For city employees, the mayor announced additional punishments. For example, City Hall workers could face suspension or be fired for owing anywhere between \$250 to \$1000 and more (Ruthhart and Reporter, 2014).

Both the Chicago police and parking enforcement agents (succinctly, non-CPD) can issue parking citations. Non-CPD agents can enforce non-moving ordinances in Chicago and are either employed directly by the city (e.g., the Department of Revenue) or through a third-party firm (e.g., SERCO). Similar to other occupations with relatively narrow objective functions (e.g., Lazear 2000), non-CPD agents believe they are evaluated based on measurable output, such as their ticketing productivity and are sometimes promoted based on their tickets per shift.[13] In contrast, Chicago police officers' main job function is not parking enforcement, and their parking ticket productivity is not as important to their careers.[14] However, ticketing revenue implicitly affects the operations of the Chicago Police Department as budget deficits can lead to citywide hiring freezes and potential layoffs (Spielman and Schuba, 2024). In addition, local officials view ticketing as a practical avenue for revenue generation (Lentino, 2017) and memos from City Hall were written when CPD-collected ticket revenue fell below targets. Subsequently CPD worked closely with the Revenue Department to track ticketing productivity (Rivera, 2010). Thus, while CPD officers may not explicitly face the same type of productivity-based incentives as non-CPD agents, they nonetheless still face top-down pressure for ticketing as a source of revenue generation. We present a simple rationalizing model that formalizes the differences between CPD and non-CPD objectives in Appendix B.

3 Data

ProPublica Illinois, in partnership with WBEZ Chicago, obtained and publicly released parking ticket data from the city of Chicago from 2007 to 2018. The data include information on the date and time of the ticket, where the vehicle was parked, the badge number of the ticketing officer, de-identified license plates, make, registration zip code, citation reason, and, importantly, payment status. This payment status includes information on the ticket

[13]This is based on reading work testimonials from Indeed.com, see `https://perma.cc/9DPN-QXF4`, where agents claim that they "work hours are late in extreme weather conditions all they care about is their ticket counts" and "all they care about is a high ticket count."

[14]Furthermore, Chicago policymakers have been shifting toward banning traffic ticket quotas for police officers in recent years. In 2019, Illinois passed a law explicitly forbidding law enforcement agencies from evaluating personnel based on their ticket-issuing productivity. This was partly due to the criticism towards CPD for mandating a minimum number of traffic stops prior to the passage of the law (Main, 2017). To the best of our knowledge, CPD officers were not required to fulfill any quotas for parking tickets and any reference to ticket quotas was in relation to traffic infractions.

outcome, such as whether the vehicle owner received a notice of seizure, whether the vehicle owner received a notice of driver's license suspension, and whether the vehicle owner declared bankruptcy as a result of the ticket. It also includes information on how much of the fine remains unpaid, the date of the last payment, and the initial fine amount.

A key object of analysis for this paper is the area or neighborhood where the ticket was given. We consider two levels of aggregation: zip code and census tract. The raw ticketing data does not include zip code or tract of violation location. To recover this information, we use the Census Geocoder, which successfully codes a vast majority of the tickets in our sample. The remainder are coded using geocoder.us. Where zip codes are unavailable from these matches, we use their latitude and longitudes to map into the relevant zip based on Census GIS shapefiles. To determine if a zip code neighborhood is considered Black or non-Black, we match each zip code to the Census 5-digit Zip Code Tabulation Area (ZCTA5) using the American Community Survey (ACS) 5-Year estimates for 2007 - 2011 (Manson et al., 2024). The ZCTA5 are approximate area representations of zip codes in the United States but are not exact matches for zip codes since ZCTA5s are aggregated from Census blocks. Despite this, they are close matches for each other. Tract demographics can be measured directly. We show in Section 5.1 that our results are robust to our choice of geographic aggregation.

Additionally, we obtained sticker registration (purchase) data by zip code using a Freedom of Information Act request filed with the City Clerk of the city of Chicago. While ideally we would be able to measure compliance at the vehicle or individual level of observation, to protect the purchasers' privacy, this information only contained the zip code of the buyer's billing address. The data contains information on the date and time of the purchase, the full purchase amount, and the type of vehicle the sticker was for. We map each sticker purchase to a ZCTA5 using the zip code.

3.1 Ticketing Sources of Revenue and Descriptive Statistics

We present summary statistics describing the ticketing data in Table 1.[15] This table reports average annual characteristics of the top ten revenue-generating tickets from 2007-2011 and characteristics for the same set of tickets from 2012-2016.[16] The type of ticket is listed in each row. We show annual ticket volume, annual ticket revenue, modal fine amount, the revenue share of the listed ticket along with the revenue share rank in Columns 5 and 11, and the ratio

[15]Displayed separately by neighborhood demographic group in Appendix Table A1.

[16]We limit the data to 2016 to ensure our estimates are uncontaminated by a series of Chicago Police Department Reforms (e.g., `https://www.chicago.gov/content/dam/city/depts/mayor/Public%20Safety%20Reforms/ProgressReport-PoliceReforms.pdf`) enacted after 2016. Our results throughout are highly similar if we use data through 2018 instead.

of revenue received to expected revenue, where expected revenue is calculated as the base fine amount times ticket volume. The revenue share is less than one when collected revenue plus applicable late fees is less than the expected collected amount if all written tickets were paid on time and is greater than one if the collected revenue plus applicable late fees exceeds this expectation. While sticker tickets were among the top two revenue generating tickets prior to the policy, they are far and away the highest revenue-generating ticket in the post-reform period, collecting around 50 percent more revenue than the next most profitable ticket. This increase in revenue is despite a sharp decline in the ratio of revenue to expected revenue, suggesting lower payment rates and increases in financial strain among the ticketed population.

Our analysis focuses on comparisons between Black (defined as $\geq 75\%$ Black) neighborhoods and non-Black neighborhoods across several ticket-related outcomes, including the type of ticket based on the violation code (sticker ticket or another type of ticket). We also include several point-in-time measures (these measures evolve over time but are current as of the latest data extract in 2019): the amount of revenue collected from the ticket, if a ticket was dismissed (either internally or as a result of an appeal) fully paid, given a notice of non-payment (if the ticket was not yet paid and the city sent a notice to the address on file for that vehicle), or included as a debt in a consumer bankruptcy case.

We show descriptive statistics summarizing the most salient data features in Table 2 for the five years before the sticker policy change (2007-2011). Panel A of this table shows information at the ticket level. About 60% of tickets in Black neighborhoods are written by CPD, who also write fewer than half of tickets in non-Black areas. About 15% of tickets written in Black areas are sticker tickets, while this rate is close to 6% in non-Black areas. Outcomes after receiving a sticker ticket are worse for people who receive tickets in Black areas as they are less likely to fully pay the ticket or have the ticket dismissed and more likely to be involved in bankruptcy or have a non-payment notice.

Panels B and C show these same outcomes at the neighborhood level. Black areas have slightly larger populations on average (about 51 thousand compared to about 47 thousand in non-Black areas) but also have fewer vehicles (about 17 thousand compared to 21 thousand). Black neighborhoods have fewer stickers purchased (16 thousand compared to 21 thousand) and slightly lower ratios of stickers purchased relative to the number of total vehicles (95% compared to 100%).[17] The number of total tickets written by the CPD is lower in Black neighborhoods, although more of them are sticker tickets. Substantially more sticker tickets

[17] Our measure of neighborhood-level vehicles is survey-based and constructed from the aggregation of several categories of survey responses, one of which is topcoded. Therefore, a strict interpretation of the number of vehicles should be made with caution.

are written in Black neighborhoods (2000 more per year).

We present the raw geographic dispersion of several key measures in Appendix Figure A1. This figure shows which zip codes are greater than 75% Black (no zip code crosses the threshold to change classifications during our sample period). It also shows the dispersion of crime (from the CPD open data portal), sticker tickets issued by CPD and non-CPD agents, and parking lots and garages in the city of Chicago (which we code based on data from Open Street Maps). The similarities between the geographies of crime and CPD-issued sticker tickets provide early evidence of the multi-tasking and effort channels discussed throughout this paper. Moreover, both CPD and non-CPD agents issue tickets throughout the city, suggesting our results below cannot be explained under alternative theories where non-CPD only work in non-Black areas or where Chicago police are not present.

4 Empirical Strategy

Our objective is to estimate the relative effects of the ticketing reform on parking enforcement patterns across neighborhoods. We summarize this difference-in-differences approach in two equations. First, in Equation (1), we present event study evidence comparing majority ($\geq 75\%$) Black ($Black_i$) areas to non-majority Black areas, where the α_t coefficients trace the evolution of an outcome of interest relative to 2011, the year prior to the sticker policy change. This allows us to carefully evaluate parallel trends, a requisite identifying assumption.

Another identifying assumption for a difference-in-differences analysis is no impact of the treatment on the control group. In our setting, changing the sticker policy may also impact outcomes in non-Black areas. What we recover from our analysis is not the overall impact of the policy on our outcomes but rather the differential impact of the policy on Black areas relative to their non-Black counterparts. Formally, for neighborhood i and year t, we estimate:

$$Y_{it} = \alpha_0 + \sum_{\substack{t=2007 \\ t\neq 2011}}^{2016} \alpha_t (Black_i \times Year_t) + \sigma_i + \lambda_t + \varepsilon_{it} \tag{1}$$

Second, we summarize our results in a single point estimate in Equation (2). Conditional on our identifying assumptions, β_1 of this equation summarizes the impact of the policy change in the follow-up period (2012-2016).

$$Y_{it} = \beta_0 + \beta_1 (Black_i \times Post_t) + \sigma_i + \lambda_t + \epsilon_{it} \tag{2}$$

A two-way fixed effects strategy is appropriate in this setting as treatment occurs simultaneously for all treated units, allowing us to avoid concerns of negative weights in the presence of treatment heterogeneity (Goodman-Bacon, 2021). Additionally, we handle the potential complications of a continuous treatment by defining a binary treatment indicator defined as ($\geq 75\%$) Black zip codes (Callaway, Goodman-Bacon, and Sant'Anna, 2021).

Our identification strategy is most credibly interpreted as generating partial equilibrium results. However, given stable and flat trends in ticketing behavior in white neighborhoods and the additional observation of little change in non-CPD behavior, the approach may credibly inform more general conclusions.

Sticker purchasing data is only available at the zip code level while ticketing data is available at both the zip and tract geographic levels. For internal consistency across exercises, we show the zip code results in the main text. In robustness checks below, we show that our results are similar when we instead define neighborhoods at the tract level. Moreover, while discretizing the treatment allows us to avoid the challenges of continuous treatment in DiD settings, the choice of treatment cutoff may be important for interpreting our results. Section 5.1 shows that our results are robust to both higher and lower thresholds.

5 Estimates of Disparate Ticketing and Outcomes

We begin our analysis by examining ticketing trends in the raw data. Panel A of Figure 1 plots the yearly number of CPD-issued sticker tickets issued in Black and non-Black neighborhoods (blue and gray respectively), both in level and per-capita terms (solid and dashed lines, respectively). Panel B, plots the same series for non-CPD agencies. In Panel A (CPD), prior to the 2012 reform, both sticker ticket series were trending downwards, with lower ticket volumes year-over-year. After 2012, however, ticket volumes in Black neighborhoods exhibited a precipitous jump upward, exceeding their pre-policy levels, while ticket volumes in non-Black neighborhoods largely flattened. In contrast, the series for non-CPD agencies (Panel B) are relatively flat, displaying little noticeable changes pre- or post-reform. Together, these raw data series depict evidence that law enforcement agencies enforced sticker ticketing more in Black neighborhoods, compared to non-Black neighborhoods.

In order to causally link the disparate enforcement to the change in sticker ticket volume and other outcomes, we must rule out other potential confounding factors. For example, neighborhoods may differ in their baseline financial strain, resulting in differential non-compliance or increased probabilities of non-payment. Neighborhoods may also experience differential policing patterns, which increase the likelihood that any non-compliance is noted by law enforcement. Our empirical strategy accounts for time-invariant differences across neighborhoods, along with common temporal shocks, although the existence of remaining time-varying differences across neighborhoods would violate the parallel trends assumption and undermine any causal interpretation.

In Figure 2, we report event-study coefficients on the interactions of neighborhood type and year indicators from Equation (1). In the plot region, we also report the summary difference-in-differences calculation, estimated using Equation (2). We begin in Panel A by examining changes in sticker ticketing volume around the time of the policy, separately for CPD and non-CPD agents. Consistent with the patterns in the raw data, we find sharp relative increases in sticker ticketing in Black neighborhoods for CPD officers (blue series). On average over the follow-up period, this increase corresponds to 2,490 additional sticker tickets per year. Reassuringly, we also find little evidence of differential pre-trend violations, lending credibility to our research design. Despite the sharp increases in relative ticketing among CPD officers, we detect no such change for non-CPD agents, and if anything, find a small decrease in relative sticker ticket volume in Black neighborhoods.

Alongside the increase in sticker ticket volume, we also find a sharp rise in relative ticket revenue collected from Black neighborhoods in Panel B. On average, CPD ticketing behavior generates an additional $270,000 per year from Black neighborhoods, which given the historical patterns of income and wealth in the city, represent a likely increase in tax regressivity. Similar to our findings in Panel A, we also detect no such relative increase in revenue from non-CPD agents.

A unique feature of our dataset is the ability to link tickets to their downstream outcomes, which we assess in Panels C through F. Among the nearly 2,500 additional annual sticker tickets in Black neighborhoods, we find that under 15% result on-time payment (Panel B) and over half receive a notice of late payment (Panel C).[18] Perhaps strikingly, almost 10% of the sticker tickets are included in bankruptcy filings. This result on relative bankruptcy is consistent with the findings in Morrison, Pang, and Uettwiller (2020), where discharging debt through Chapter 13 bankruptcy filings permit individuals to retain access to their car and driver's license, a benefit which accrues more strongly to Black residents. This rise in bankruptcy filings may also reflect differences in underlying financial strain in the population, where the marginal sticker ticket has a disparate impact on the financial health of Black versus non-Black residents. Finally, we observe some rise in dismissed sticker tickets in Black neighborhoods, which are likely due to ex-post changes in dismissal policy and debt relief programs (e.g., Sanchez and Ramos 2015), rather than evidence on the ticket quality of the marginal sticker ticket.[19]

[18]Tickets can also accrue revenue through late penalties and interest, even if they ultimately end up in bankruptcy or have a late-payment notice in our data.

[19]In Appendix Figure A2 we re-estimate Equation (1) at the ticket-level. The marginal CPD-written ticket is 3.8 percentage points more likely to be a sticker ticket. Conditional on a sticker ticket, the marginal ticket

It's possible some of the observed disparity is due to changes in neighborhood differences in resident's ability to pay for the sticker itself, resulting in differential non-compliance with the policy. Consequently, individuals who are unable to initially afford the sticker will also be unresponsive to the purchasing incentive induced by the sticker fine increase or be priced out because of the increase in the sticker price itself. Recall that our DiD estimates account for the initial level disparity across neighborhood types, and thus capture only the differential change in sticker ticketing frequency across neighborhoods, before and after the reform. However, we take seriously the idea of quantifying differential compliance and its interaction with departmental incentives as a mechanism for our results and discuss this point in detail in Section 5.2.

5.1 Robustness Checks

Before assessing the mechanisms underlying our results, we first examine the stability of our estimates to a range of robustness checks in Appendix Table A2. Column 1 in each table reports the DiD estimate for each of our six main outcomes using only the raw data. Column 2 reproduces our estimate from the main text, adding zip code and year fixed effects in a standard DiD specification, although the results are little changed with these additions. Column 3 shows that our estimates are virtually unchanged when using the estimator suggested by Callaway and Sant'Anna (2021). Finally, in the last two columns, we use alternative cutoffs for defining zip codes as primarily Black. Changing the threshold to either at least 50% or 90% Black, rather than our baseline 75%, does not meaningfully affect our findings.[20]

We also decompose the main analysis for the subset of tickets that have owner characteristics in Appendix Table A3 to determine whether the disparate ticketing patterns documented above largely accrue to individuals whose home neighborhood matches the racial composition of the ticketing zip code or if the results we find largely reflect commuter traffic instead. The first row of each panel ("main") reproduces the difference-in-differences estimates from Figure 2. We then replace each outcome Y_{it} with $Y_{it} \times (Black\,Owner_i)$ in the second row and $Y_{it} \times (1 - Black\,Owner_i)$ in the third row, effectively decomposing the differential outcomes in Black neighborhoods to drivers from Black and non-Black neighborhoods, respectively.[21] Across nearly all outcomes, regressions, and ticketing agencies, we find that the burden of

collects $27 fewer in revenue in Black neighborhoods, is more likely to end in a notice of non-payment and end in bankruptcy.

[20]Appendix Figure A3 finds broadly similar results but replaces the zip code fixed effects with tract fixed effects and redefines neighborhoods as majority-Black at the tract level using the same 75% threshold.

[21]We note that since not all of our tickets contain owner address information, the sum of the two disaggregated point estimates need not add up to the main results.

the disparate ticketing patterns in Black neighborhoods tends to fall on owners who are also from Black neighborhoods.

5.2 Mechanisms: Search Costs (Not Compliance) Drive Results

CPD and non-CPD agents perform different tasks and face different incentives. CPD officers have a broad set of responsibilities, including public safety priorities along with ticketing, while non-CPD agents are more specialized in ticketing. While the two groups seem to have reached an equilibrium prior to the ticketing reform, the incentives and responsibilities faced by the two agencies imply, and we empirically find, differing responses to a top-down push for increased revenue, with CPD agents responding more strongly to these incentives (see Appendix B for a formalized model). This subsection explores two potential explanations for the varied responses: changes in agency enforcement and resident compliance and evasion. We present evidence that CPD agents respond to their specific incentive structure, and in particular the push for additional revenue, by increasing enforcement in Black neighborhoods, likely because they are multi-tasking from other policing activities occurring in Black neighborhoods. That is, the marginal search cost may be relatively lower in these neighborhoods. Non-CPD agents do not have similar revenue incentives, nor do they have policing activities which reduce marginal search costs in Black neighborhoods, and thus do not respond to the policy with disparate enforcement. Finally we rule out a differential compliance story, as we do not find evidence of relative changes in sticker purchases across Black and non-Black neighborhoods.

Departmental Incentives and Search Costs: An implication of the different responses across ticket-writing departments is that the underlying performance evaluation scheme induces differential responses to the policy. Since non-CPD agents are evaluated on their ticket volume, their ticket-writing behavior was already maximizing ticket volume while minimizing search costs. Changing the revenue from a given ticket leaves the volume problem unchanged. In contrast, CPD officers face no such volume-based incentives to our knowledge and care more about revenue.[22] Thus, post policy change, the marginal benefit of writing an additional sticker ticket, from a revenue collection standpoint, has increased for CPD only. As a result, officers may induce greater search efforts into finding or ticketing vehicles without appropriate city stickers while multitasking on their other responsibilities.

We partially test whether officers exert greater search effort into finding vehicles with expired stickers across all areas by plotting the number of sticker tickets issued by day for 2011 and 2012 in Figure 3. If officers exert greater search effort when the revenue benefit of

[22]See Section 2 for additional discussion on ticket volume and revenue incentives across agencies.

doing so is greater, then we would expect to see increases in ticketing frequency immediately after the 15-day sticker renewal grace period ends and tickets can be legally issued. In Panel A, we see exactly this pattern for CPD, with a large spike in sticker tickets written on the day the grace period ends. Comparing the change between 2011 and 2012, sticker ticket volume increased by a dramatic 49.5% on the day immediately following the grace period.

In contrast, non-CPD agencies (Panel B) exhibit only a 17.0% increase in sticker ticket volume.[23] Interestingly, we also see small ticket volumes in the days preceding the end of the grace period, although an expired sticker should technically not be subject to enforcement in this window. The share of sticker tickets written in the expiration-grace period window is smaller for non-CPD agencies (<1%) than CPD agencies (between 4.9 - 5.6%), which we take as suggestive evidence of ex-ante ticket writing optimization by the former. The same histograms for non-sticker tickets have little evidence of similar discontinuous behavior.

We provide additional evidence CPD officers are responsive to revenue collection objectives using the sale of Chicago parking meters to private investors in December of 2008. Under the privatization agreement, the city of Chicago would receive a one-time payment of $1.15 billion from private investors in exchange for the next 75 years of parking meter revenue. Since the revenue from parking meter tickets no longer flows to the city budget after the sale, CPD officers should be less likely to write meter tickets, relative to non-CPD contractors, if CPD ticketing patterns are responsive to revenue incentives. We test this hypothesis in Appendix Figure A4 and find sharp relative decreases in CPD-written meter tickets after meter privatization, consistent with CPD officers altering their ticketing behavior in response to the destination of the revenue.[24]

Next, we test how sticker ticket enforcement patterns correlate with alternative neighborhood characteristics. We replace our primary $Black_i \times Post_t$ interaction with different interactions based on pre-reform neighborhood characteristics in Appendix Table A4. If officers are behaving in a purely revenue-maximizing way (without incorporating search costs), then sticker tickets should be written in the areas that have the highest repayment probabilities, such as high-income neighborhoods.[25] In fact, we find the opposite patterns in Column

[23] The cyclical pattern in non-CPD ticket volume is a weekend/weekday effect. Therefore, we compare ticket volume on the first weekday after the grace period in 2011 (two days after it ends) against the initial expiration day in 2012 in the above calculation.

[24] Consistent with general revenue concerns being a key motivator, we also find suggestive evidence of relative increases in non-sticker ticketing in Appendix Figure A5, however these estimates are partially based on unstable pre-trends.

[25] It's ambiguous whether officers would fully internalize potential ticket contesting or repayment probabilities when comparing the value of a ticket in high and low-income neighborhoods. Higher-income neighborhoods likely have more resources to fight parking tickets, but to the extent officers receive overtime pay for appearing in court (e.g., Chalfin and Goncalves 2021), the marginal ticket in a high-income neighborhood becomes attractive both with respect to repayment probabilities as well as potential private value to the

2.

We also examine the relationship between other policing activities and ticketing patterns for CPD officers. In Column 3, we define crime as the total crimes per 10,000 residents, and the "high crime" indicator includes the zip codes in the upper quartile. The results suggest at least a part of the story is that officers, often located in high-crime areas, begin spending more of their time issuing sticker tickets. In Column 4, we show that CPD officers write more sticker tickets in areas that previously had high rates of sticker tickets (as a fraction of total tickets written in the neighborhood), suggesting that officers are aware of neighborhoods with low sticker compliance rates and alter their search effort accordingly. However, Column 5 shows that fewer sticker tickets are written in neighborhoods with high payment rates. When we test all interactions jointly, we still find that while the coefficient on neighborhood demographics remains large, disparities seem to also be significantly associated with area crime characteristics, indicating that CPD officers increase ticketing behavior in areas where they are already located due to other responsibilities.[26] Alternative non-race and non-crime channels are less important.[27]

Finally, we further emphasize the complementary role of multitasking and its interaction with revenue collection incentives in Panel A of Figure 4, where we plot the joint distribution of changes in CPD-issued sticker tickets against the average annual number of crimes at the neighborhood level.[28] Consistent with multitasking crime response efforts reducing the marginal cost of ticketing, we find the strongest ticketing responses in areas with the greatest amount of criminal activity, which subsequently concentrates ticketing in Black neighborhoods. In stark contrast, the same correlation for non-CPD agents is flat, indicating no relationship between a neighborhood's underlying crime level and the issuing of sticker tickets.[29] Taken together, we conclude that both top down revenue pressures rather than volume-based incentives and the differences in multi- vs uni-dimensional responsibilities

officer.

[26]It is worth noting the tension of this finding with how CPD's union describes the multi-tasking problem: "fewer tickets are written in higher-crime areas because officers are busy chasing more serious offenses." `https://abc7chicago.com/archive/7614129/`

[27]We further test for differential officer behavior *within* Black neighborhoods by examining all alternative treatment margins above, but defined within the set of high-Black neighborhoods, as opposed to the overall sample in Appendix Table A5. Within Black neighborhoods, we find evidence that sticker tickets are weakly more likely to be written in higher-income neighborhoods and in areas with high baseline sticker ticket rates. Consistent with the previous cross-neighborhood exercise, non-CPD agents exhibit little differential enforcement across subsets of Black neighborhoods.

[28]Formally, we consider all one-way differences for $i \in \mathcal{I}$ and $R_i \in \{B, n\}$ of $E[Y_{it}|R_i = r, Post_t = 1] - E[Y_{it}|R_i = r, Post_t = 0]$, which represent each component piece of our difference-in-differences estimates. Averaging across all of these estimates by neighborhood type R_i and then taking the difference recovers our main DiD estimates.

[29]Reassuringly, we also find no differential changes in crime or police activity across neighborhoods around the time of the reform (see Appendix Figure A6).

across agencies are key drivers in determining the disparate responses we document above.

Differential Compliance and Evasion: Concerns over tax evasion have a rich history in the public finance literature (Allingham and Sandmo 1972; Slemrod and Yitzhaki 2002), and one interpretation of our existing estimates is that they simply reflect differences in the willingness or ability of drivers to pay for the city sticker or evade enforcement efforts. As described above, differential ex-ante compliance with the policy may present itself as disparate ex-post enforcement of the sticker tax if the marginal benefit of writing such a ticket has increased, such that law enforcement agencies are now writing tickets they would not otherwise have in the absence of the budget reform. Alternatively, the 2012 budget reform may have changed compliance rates themselves since it also increased the price of the sticker for both small ($75 to $85) and large ($120 to $135) vehicles. If there is a sufficiently large subpopulation on the margin of sticker purchasing, then such an increase may lead to disparate enforcement as the marginal search cost of finding a delinquent motorist has relatively decreased in areas with larger decreases (or smaller increases) of sticker purchases. While we are unable to directly measure sticker purchases at the individual level due to data limitations, we conduct several tests to probe how much our estimates may reflect differential compliance versus differential enforcement of the policy.

First, we examine neighborhood-level sticker purchasing behavior using administrative data on sticker purchases, owner locations, and sticker types from 2008-2016. In Panel A of Appendix Figure A7, we plot event-study estimates of the interaction of year and neighborhood-type indicators, using sticker purchases as the outcome. If there were differential non-compliance with the sticker policy, such that the fine increase induced a large fraction of the non-complying population to suddenly purchase tickets, then we should see greater purchase rates in Black neighborhoods relative to non-Black neighborhoods. Alternatively, the sticker price increase may also lead to a differential reduction in purchasing as marginal individuals are priced out of compliance. If anything, Black neighborhoods have minuscule decreases (38 stickers) relative to non-Black neighborhoods, although the pre-trend estimates are somewhat noisy. However, in aggregate, both groups increased their purchases. When we disaggregate sticker purchases into types focusing on passenger vehicles, we find no statistically distinguishable differential response. This empirical finding is instead consistent with changes in officer behavior rather than substantial changes in civilian behavior due to price-out non-compliance or incentivized purchasing.[30]

Car owners may also alter their parking behavior in an attempt to evade enforcement.

[30]We confirm this finding through a decomposition exercise in Appendix Table A6 where we relate the magnitude of the enforcement and compliance responses across neighborhoods, adjusting for the presence of unregistered vehicles. We continue to find strong differences in enforcement responses across neighborhoods alongside no detectable differences in compliance behavior or stark gaps in the stock of unregistered vehicles.

Enforcement may vary based on parking location.[31] We can (with some noise) measure whether a ticket was issued in a parking lot or structure. To do so, we measure whether a sticker ticket was issued at a location classified by Open Street Maps as a parking amenity.[32] 3.3% of CPD sticker tickets are issued in parking amenities (4.1% in non-Black zip codes, 2.2% in Black). 4.5% of non-CPD sticker tickets are issued in parking amenities (5.3% in non-Black zip codes, 1.6% in Black). We map this in Appendix Figure A1.

While not a perfect test for car owner behavior, Appendix Figure A8 shows that the share of tickets issued by CPD in parking areas does not substantially change in Black zip codes relative to non-Black zip codes after the policy change. This result is evidence that drivers in non-Black areas are not changing their parking behavior in order to avoid more costly enforcement and instead is consistent with increased search effort from CPD.

Is Differential Enforcement Maximizing Revenue?: Black neighborhoods receive substantially greater levels of sticker ticket enforcement, in magnitudes that cannot be rationalized by changes in sticker purchasing behavior. And moreover, this disparate effect is entirely driven by CPD rather than non-CPD ticketing agencies. One possible rationale for these empirical patterns is that CPD officers are differentially targeting areas with lower sticker purchase rates or higher repayment rates, while non-CPD officers have already equalized their marginal costs of enforcement across neighborhoods. To understand how neighborhood responses and baseline characteristics influence enforcement behavior, we construct joint distributions of estimated changes in neighborhood-level ticketing outcomes and baseline neighborhood characteristics.

In Panels A and B of Figure 5, we plot the joint distributions of neighborhood-level changes in sticker tickets and sticker purchases, separately by neighborhood type and ticket issuing agency. There is a weakly positive correlation between changes in sticker purchases and CPD-issued sticker tickets in non-Black neighborhoods, which is sharply contrasted with a negative relationship in Black neighborhoods (Panel A). Strikingly, non-CPD-issued sticker tickets display parallel and virtually flat relationships across neighborhood types (Panel B). A possible interpretation of the negative slope is that CPD agents are more successfully capturing every marginal change in sticker purchases; perfect enforcement would suggest a

[31] To the extent vehicles in Black neighborhoods are more likely to be parked on the street or visible to the average patrol (e.g., Sanchez and Ramos 2018), then Black neighborhoods may see differential levels of sticker enforcement even in the pre-reform period since Black neighborhoods exhibit a lower marginal search cost for sticker-less vehicles.

[32] See `https://wiki.openstreetmap.org/wiki/Tag:amenity=parking` for details on how Open Street Map classifies parking amenities. Mapping tickets into parking areas is done with some noise, as the ticket latitude and longitude are often for the street address rather than within the parking area. Thus, we impose a buffer, and a ticket is deemed "in a parking amenity" if it occurs within 15 meters from a parking amenity or within a parking amenity as defined by Open Street Mapping. Results are robust to alternative buffer regions; even at a buffer of 30m, fewer than 9% of sticker tickets are in parking amenities.

slope of -1. We estimate a slope coefficient of -0.52. However, this slope difference does not fully explain the level difference between Black and non-Black neighborhoods with similar purchase responses, suggesting alternative mechanisms such as search costs.

We further explore whether differences in neighborhood characteristics can explain the CPD ticketing gap in Appendix Figure A9. In Panel A, we test whether differential pre-reform sticker purchase rates can rationalize such a gap, but we continue to find a persistent level difference, with virtually all neighborhood purchase rates clustered closely around one.[33] Alternatively, such a disparity may be justified by revenue collection motivations if the repayment probabilities are higher in Black versus non-Black neighborhoods. However we find that sticker ticket volumes are higher in Black neighborhoods, despite having lower pre-reform sticker ticket payment probabilities (Panel B). Finally, we show that changes in ticket volumes remain higher in Black neighborhoods across almost all values of sticker tickets and paid sticker tickets per sticker purchase (Panels C and D), indicating sharply distinct enforcement responses across areas with similar baseline enforcement propensities.

Taken together, we conclude that observable differences in neighborhood characteristics, compliance, and expected payment probabilities are insufficient to fully explain the CPD sticker ticketing gap across Black and non-Black neighborhoods. Instead, our results are consistent with a differential response to the fine increase by ticketing agency, combined with differential marginal search costs across neighborhoods. Therefore, the interaction of departmental incentives and search costs together plays a key role in determining disparate enforcement patterns across agencies.

Given the overlap in the support of non-race neighborhood characteristics, there are potentially significant revenue gains from equalizing enforcement across Black and non-Black neighborhoods. A back-of-the-envelope calculation leveraging the joint distributions of neighborhood characteristics and neighborhood-level ticketing responses suggests that the city could have raised an additional $1.9 million in annual revenue by applying average equal enforcement in low-compliance non-Black neighborhoods.[34]

[33] Some of the x-axis dispersion is likely due to measurement error in the number of vehicles since we rely on aggregated survey-based measures when constructing this statistic.

[34] To perform this calculation, we first identify non-Black neighborhoods with lower purchase per vehicle rates than the minimum rate of Black neighborhoods. Intuitively, these are the neighborhoods where non-compliance is highest, and the marginal cost of enforcement is consequently lowest (abstracting from parking patterns). We then assign the mean Black neighborhood enforcement volume to these neighborhoods and assume the payment probability is equal to the pre-period payment rate. The mean payment rate for these focal non-Black neighborhoods is 59.5%, relative to 46.9% in Black neighborhoods.

6 Estimating Officer-Specific Responses to the Reform

Our aggregate event study results reveal disproportionate issuance of sticker tickets across neighborhoods. An open question is whether this disparate behavior is department-wide or if it is concentrated among a handful of officers who are high-volume ticket-writers.[35] We estimate a modified version of our difference-in-differences specification above to decompose the response to the policy reform across the officer distribution. Formally, we estimate:

$$Y_{ijlt} = \underbrace{\sum_{j \in J} \delta_j (Z_j \times Post_{lt})}_{\text{Officer-specific responses}} + \underbrace{Z_j}_{\substack{\text{Officer fixed} \\ \text{effect}}} + \mathbf{X}'_{it}\pi + \nu_{ijlt} \tag{3}$$

for neighborhood i, officer j, ticket l, and year t. We control for year, unit, neighborhood, and officer fixed effects.[36] We study racial differences in officer ticketing by interacting Y_{ijlt} with indicators for neighborhood racial composition instead of placing the interaction term on the right-hand side.

The coefficients of interest are the interactions δ_j, which, conditional on the officer fixed effects, nonparametrically capture the within-officer response to the change in incentives induced by the fine increase. When estimated at the sticker ticket level, δ_j can be interpreted as the officer-specific outcome of the marginal ticket written in response to the policy.[37]

6.1 Decomposing Outcomes of Marginal Sticker Tickets

Figure 6 reports the distributions of officer-specific policy responses. In Panel A, we plot bins of the officer-specific probability of writing a sticker ticket separately for Black and non-Black neighborhoods against the overall officer-specific sticker ticket response ($P(sticker|ticket, race)$ against $P(sticker \mid ticket)$). We report similar findings from Empirical Bayes-adjusted estimates in Appendix Figure A10. This exercise effectively decomposes the marginal sticker

[35] Our ticketing data includes officer badge numbers. Because these identifiers can change, we build an officer badge crosswalk using data from OpenOversight. We combine this with data from the Invisible Institute to generate an officer-level data set with information on the number and type of tickets written, unit and employment history, complaints against the officer, and officer demographic information (race/ethnicity, sex, and age).

[36] These fixed effects account for time-invariant differences in responsibilities and ticket writing potential across both unit assignments and geography. We depart slightly from our aggregate analysis and use tract (rather than zip) fixed effects and racial composition to more closely approximate officer beat assignments. These modifications allow us to identify racial differences in policy response within all officers rather than only among the subset of officers whose assignment happens to be near zip code boundaries.

[37] We restrict the sample to officers who write at least 100 tickets in our sample period and write tickets both before and after 2012 to alleviate noise concerns. These restrictions drop only a handful of officers. Unfortunately, we cannot conduct a similar exercise for non-CPD units due to higher rates of turnover in those departments.

ticket response outcomes within each officer. Comparing the slopes of the race-specific against the overall response yields similar conclusions to our aggregate event studies above - officers responding to the policy reform are substantially more likely to write sticker tickets in Black neighborhoods than non-Black neighborhoods (0.641 vs 0.359). The magnitude of the overall sticker ticket response on the x-axis also provides evidence of a "first-stage" response to the policy, as greater than 65% of officers are more likely to write a sticker ticket post-reform.[38]

Panel B examines the officer-specific revenue responses accruing from payment sticker tickets. The correlations between the race-specific and overall responses suggest that over 60% of the revenue originates from Black neighborhoods. Together with our decomposition results, these correlations suggest that the unequal enforcement of the sticker ticket also led to an unequal tax burden across neighborhoods.

In Panels C and D, we decompose the outcomes of the marginal sticker ticket separately for Black (Panel C) and non-Black (Panel D) neighborhoods. Specifically, we plot the correlations between $P(ticket\ outcome \mid sticker, race)$ against $P(sticker \mid race)$. There are striking differences in sticker ticket outcomes across race. Under 35% of sticker tickets written in Black neighborhoods end in payment, compared to non-Black sticker tickets, which are almost 15 p.p. more likely to end in payment. Sticker tickets in Black neighborhoods are significantly more likely to end in financial strain, consistent with our aggregate results above. Fully 45.3% of sticker tickets either receive a notice of non-payment (39.7%) or end in the driver filing for bankruptcy (5.6%). While sticker tickets in non-Black neighborhoods also only end in payment about half the time, there are fewer adverse financial outcomes (23.2% notice, 1.3% bankruptcy). Taken together, our results align with our aggregate analysis that the marginal sticker ticket is more likely to occur in Black neighborhoods, disproportionately generates revenue from this population, and leads to substantially worse financial outcomes compared to non-Black neighborhoods, and that these effects are widespread across the officer distribution.[39]

6.2 Correlating Policy Responses with Officer-Level Observables

Finally, we explore whether the magnitude of the disparate sticker response is correlated with officer-level observables. While the majority of officer responses indicate that they are

[38] A version of this exercise which captures both the extensive and intensive margin, reveals that over 85% of officers are responsive in a "first-stage" sense, providing additional support of behavioral changes to the policy reform across the officer distribution.

[39] The disparate marginal revenue result in Panel B despite the lower payment rate in Panel C is likely driven by increases by racial differences in accrual and payment of late fees and extra penalties. Regressing the revenue received against the paid ticket probabilities reveals that the marginal paid Black sticker ticket generates $278 of revenue compared to $236 from non-Black sticker tickets.

responsive to the sticker ticket fine increase in a first-stage sense, understanding whether officer characteristics are predictive of the magnitude of observed responses is important for characterizing how different types of officers react to incentive changes, as well as for designing department-level policies which may mitigate such adverse incentive responses.

A simple explanation for our officer-level decomposition is that the responses reflect the demographics of the unit that they are assigned to. That is, officers assigned to units with greater Black population shares should also exhibit larger Black ticketing responses. Conversely, officers assigned to units with smaller Black population shares should exhibit larger non-Black ticketing responses. We test whether this mechanism drives our results in Appendix Figure A11, regressing race-specific δ_j responses against their modal unit assignment's Black share of the population.[40] Consistent with neighborhood demographics playing a key role in determining the responses in Black neighborhoods, we find a strong, positive correlation between estimates of $\delta_j(Sticker, Black)$ and neighborhood demographics. In sharp contrast, however, we find a muted correlation between estimates of $\delta_j(Sticker, Non{-}Black)$ and neighborhood demographics, rather than a negative correlation, which a pure neighborhood characteristics story would predict.

Given this stark contrast, we next examine whether officer observables predict their policy responses in Table 3, controlling for modal unit fixed effects throughout.[41] In each successive column, we test a series of covariates before pooling them all together in Column 4. Black officers consistently have smaller $\delta_j(Sticker, Black)$ responses, along with more experienced officers, though this latter correlation dissipates in Column 4. In contrast, officer characteristics are generally uncorrelated with $\delta_j(Sticker, Non - Black)$, with the exception of ticket volume. These estimates suggest some degree of differential leniency or search effort based on the interaction of officer race and neighborhood demographics. The magnitudes of the δ_j responses in Black neighborhoods are declining with experience, which is also indicative of early career officers perhaps being more responsive to revenue collection efforts in ways that disparately impact the population.

More generally, the combination of a strong policy response across the officer distribution and weak correlations with observable characteristics suggests that the empirical patterns we find in this paper result from a broader goal of revenue generation and multitasking responsibilities on a department-wide level. Given the disparate impacts in the population that clearly hinge on ticketing agency, revenue collection as one responsibility of law enforcement agencies may benefit from specialization.

[40] We restrict the sample to police officers who are in patrol units so that we can correctly estimate the Black population share in the assignment, as well as examine responses of police officers who are most likely affected by the policy.

[41] We show the correlations using the complete set of officers in Appendix Table A7.

7 Conclusion

In this paper, we examine the role of policing on the distribution of tax burden on residents by exploiting the fine increase for vehicle registration non-compliance in Chicago. Using this sharp change in 2012, we showed that enforcement of this fine was indeed disproportionately distributed in the population, with Black neighborhoods experiencing far greater changes in their ticket volumes than non-Black neighborhoods.

Across enforcement agencies, we only find significant evidence of disparate enforcement when examining the ticketing behavior of police officers in the Chicago Police Department and not from tickets issued by parking enforcement agents. We show that CPD's disparate response patterns cannot be fully explained by differences in baseline non-race neighborhood characteristics, nor justified on collection revenue generation grounds or differential non-compliance. Instead, we provide evidence that the combination of a multi-dimensional objective function (and higher crime rates in Black neighborhoods) with differential marginal enforcement costs by neighborhood drive these disparate responses. A top-down focus on revenue production only serves to exacerbate these enforcement behaviors.

Together, our results provide evidence that revenue generation in local governments may benefit from specialization across collection agencies as a mechanism to mitigate disparate impacts in the local population. While parking tickets, particularly sticker tickets, currently function as a form of regressive tax, cities can implement a more equitable and revenue-generating ticketing regime by altering the incentives of the ticketing agents or by shifting parking enforcement responsibility to only parking enforcement agents. Either of these could simultaneously achieve more equitable outcomes while also raising additional revenue.

Finally, our results document preliminary evidence of a direct relationship between disparate policing and downstream financial consequences. Notably, nearly 10% of the additional sticker tickets in our setting were unpaid and included as part of bankruptcy filings. Thus, policies addressing disparate policing behavior may also reduce racial disparities in socioeconomic outcomes.

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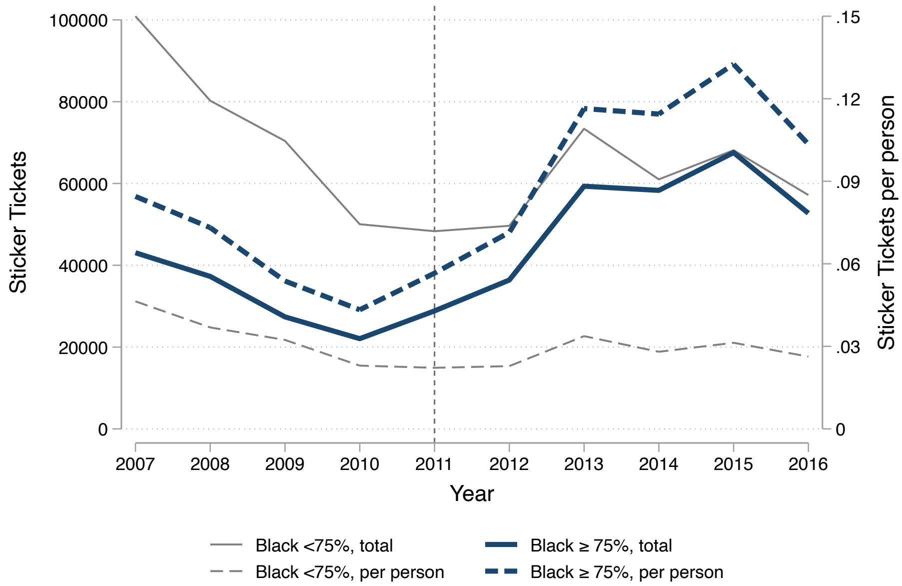
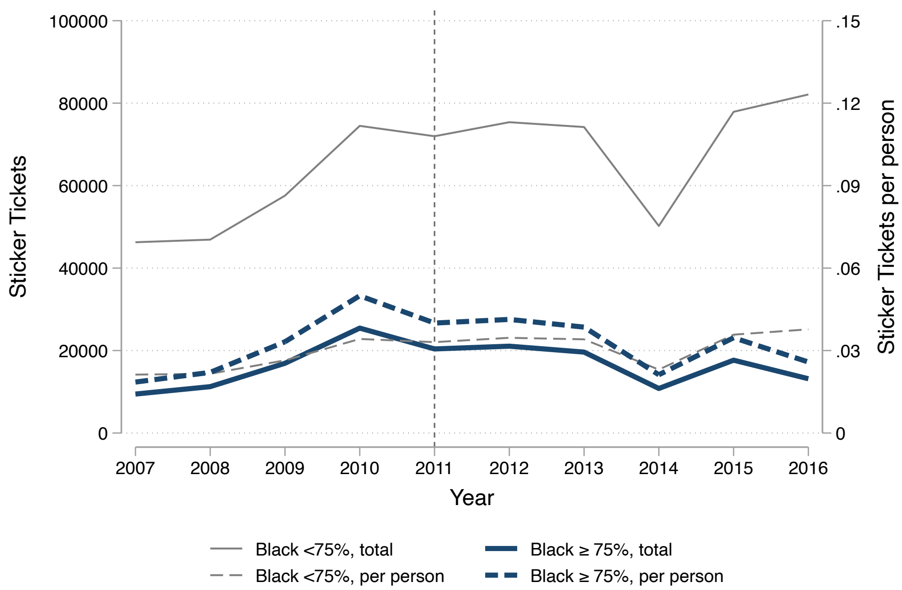
Figure 1: Time Series of Sticker Ticket Volume by Issuing Agency

Panel A: Chicago Police Department

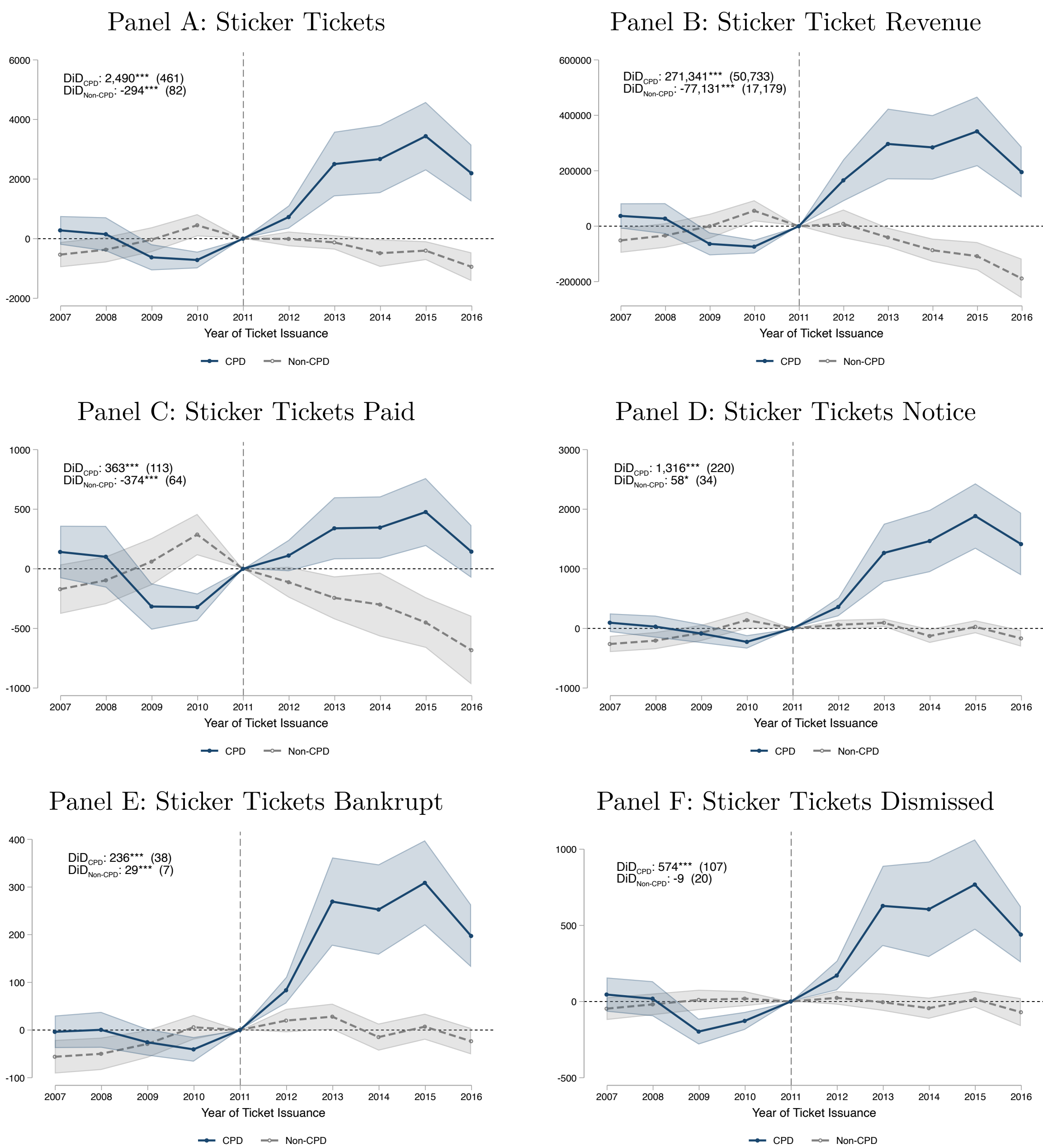


Panel B: Non-Chicago Police Department



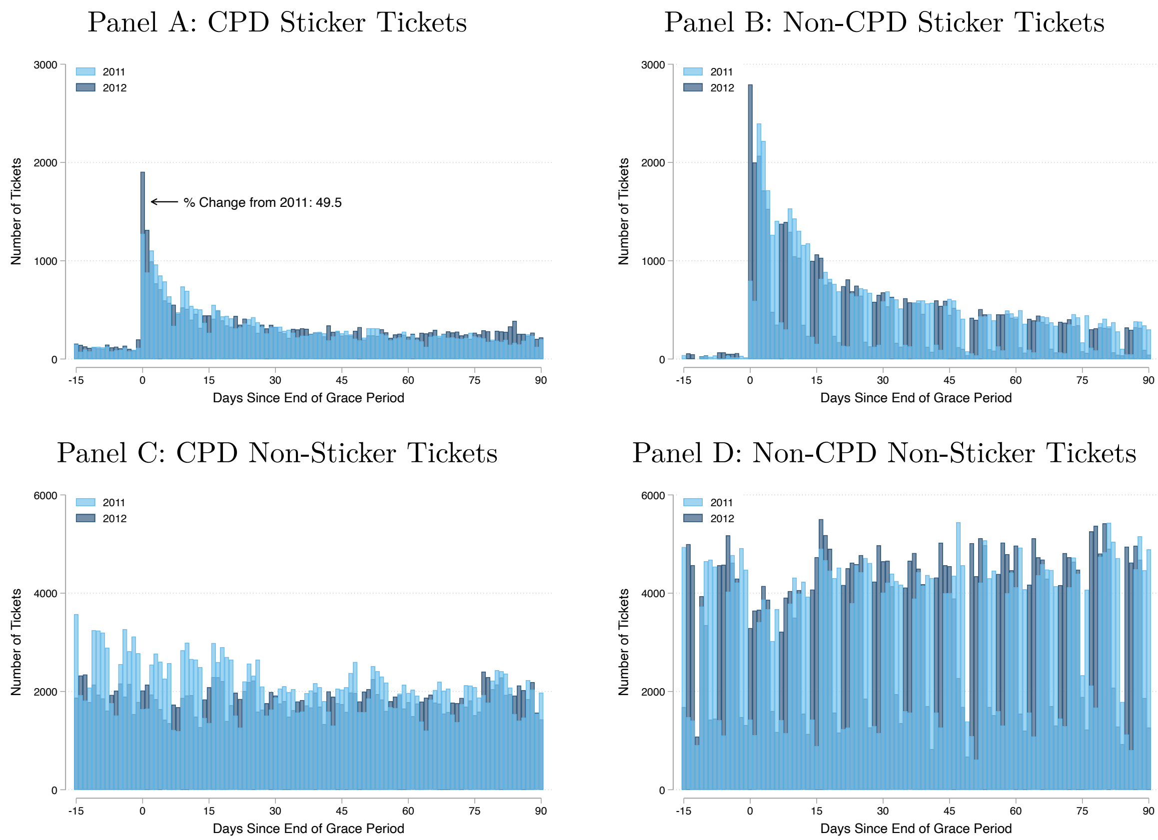
Notes: This figure reports time series of sticker tickets issued and sticker tickets issued per capita by neighborhood and ticketing agency. Black neighborhoods are defined as zip codes with greater than seventy-five percent Black population share. Panel A reports results for CPD and Panel B reports results for non-CPD agencies. Solid lines report levels (left axis), dashed lines report per-capita population rates (right axis). Blue lines represent Black neighborhoods and gray lines represent non-Black neighborhoods. The vertical line in 2011 denotes the last year prior to the reform. Population is measured using the 2007-2011 American Community Survey.

Figure 2: Event Study Estimates of Sticker Tickets and Outcomes at the Neighborhood-Level by Issuing Agency



Notes: This figure reports event study estimates of sticker ticketing behavior and sticker ticket outcomes at the neighborhood level, estimated separately for the Chicago Police Department (CPD) and non-CPD agencies. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. The blue points report estimates for CPD and the gray points represent estimates for non-CPD. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level.

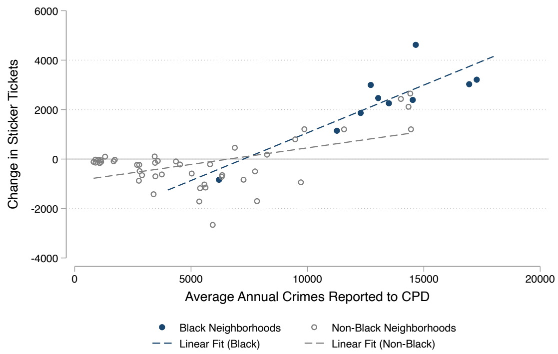
Figure 3: Distribution of Tickets Around the End of Sticker Renewal Grace Period: 2011-2012



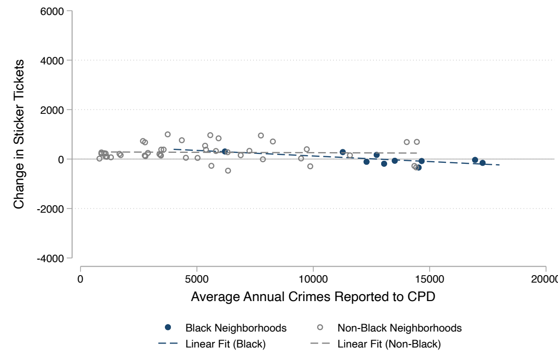
Notes: This figure reports the number of tickets issued per day in 2011 and 2012 by CPD and non-CPD ticketing agencies. Light blue histograms represent 2011 and dark blue histograms represent 2012. The x-axis is normalized to the end of the year-specific sticker renewal grace period and includes 15 days before and 90 days after the end of the grace period. Panels A and C ticketing distributions for CPD and Panels B and D report ticketing distributions for non-CPD agencies. The upper panels report the distribution of sticker tickets and the lower panels report the distribution of non-sticker tickets. The grace period is a window after the expiration date where an individual may purchase a city sticker without paying late fees and is not supposed to be subject to sticker ticket enforcement. We use 2011-2012 as the focal years in this exercise since, prior to 2014, all city stickers expired at the end of June in any given calendar year. The city shifted to time-varying city sticker expiration dates in 2014 when city stickers then expired 6 months after state-level car registration. Moreover, both 2011 and 2012 have the same fifteen-day grace period, which enables us to more cleanly harmonize and compare the data across years.

Figure 4: Neighborhoods with Higher Crime Experience More CPD Ticketing

Panel A: Δ Tickets and Crime - CPD



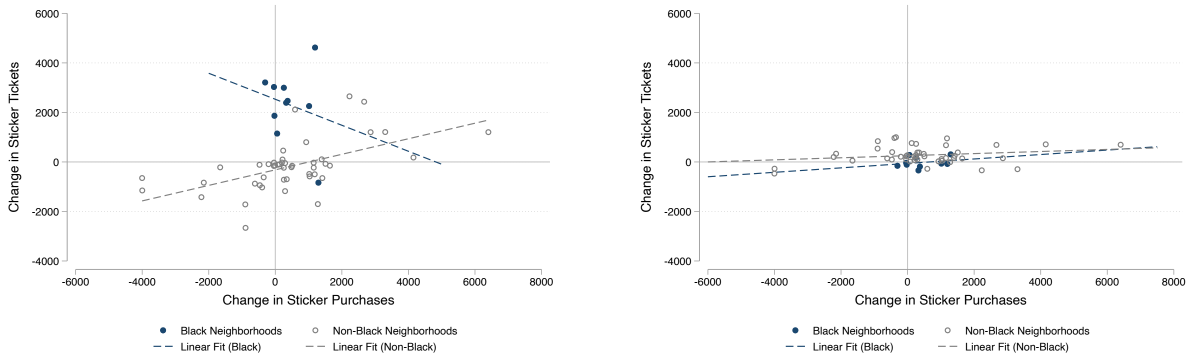
Panel B: Δ Tickets and Crime - Non-CPD



Notes: Notes: This figure reports joint distributions of neighborhood-level one-way difference estimates of the change in sticker ticketing, along with neighborhood-level estimates with pre-reform crime levels. Each panel plots the change in sticker tickets and neighborhood crime in levels for Black and non-Black neighborhoods, by CPD (Panel A) and non-CPD (Panel B) ticketing agency, respectively. Each point represents a neighborhood, defined at the zip code level. Navy dots represent Black neighborhoods and gray circles represent non-Black neighborhoods. Dashed lines represent linear lines of best fit, estimated separately by neighborhood type.

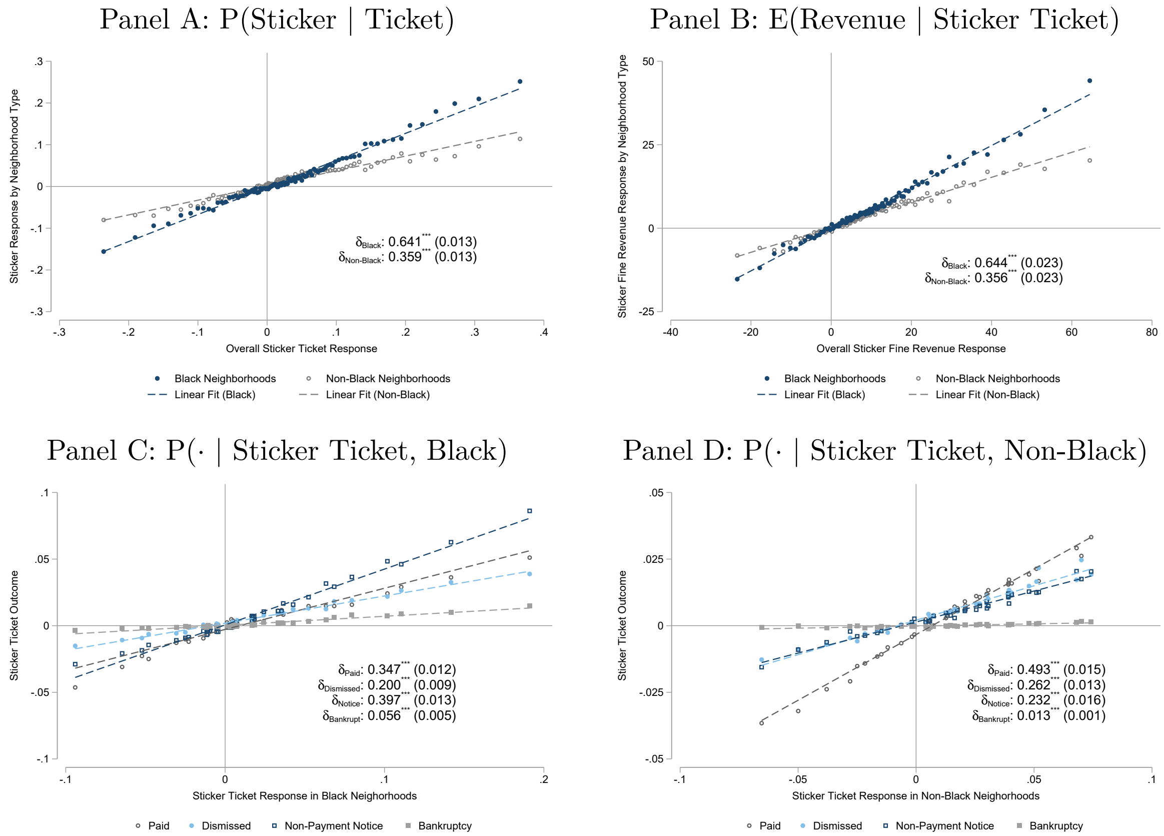
Figure 5: Relationship Between Change in Sticker Ticketing and Sticker Purchasing

Panel A: Δ Tickets and Δ Purchases - CPD   Panel B: Δ Tickets and Δ Purchases - Non-CPD



Notes: This figure reports joint distributions of neighborhood-level one-way difference estimates across different outcomes. Each panel plots the change in sticker tickets and sticker purchases for Black and non-Black neighborhoods, by CPD (Panel A) and non-CPD (Panel B) ticketing agency, respectively. Each point represents a neighborhood, defined at the zip code level. Navy dots represent Black neighborhoods and gray circles represent non-Black neighborhoods. Dashed lines represent linear lines of best fit, estimated separately by neighborhood type.

Figure 6: Estimating and Decomposing Officer-Specific Responses to Sticker Fine Increase



Notes: This figure plots estimates of δ_j for different outcomes against each other. In Panels A and B we plot estimates of neighborhood race-specific δ_j responses against the overall race-agnostic δ_j response on the x-axis. In Panels C and D, we plot the race-specific sticker ticket outcomes against the race-specific sticker ticket responses, separately by race. There are 100 bins per outcome in the top two panels and 40 bins per outcome in the lower two panels. For exposition, we drop the first and last bin for each outcome. Reported coefficients estimated on the underlying officer-level estimates. Robust standard errors are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Table 1: Annual Volume and Revenue from Top Ten Revenue Generating Tickets in Chicago

	2007-2011						2012-2016					
	Tickets (000s)	Revenue ($000s)	Avg. Fine	Rev. Share	Rev. Rank	Rev./ E[Rev.]	Tickets (000s)	Revenue ($000s)	Avg. Fine	Rev. Share	Rev. Rank	Rev./ E[Rev.]
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Expired Meter	404	19,533	50	0.14	1	0.97	283	15,265	50	0.10	4	1.08
No City Sticker	178	18,614	120	0.14	2	0.87	205	29,575	200	0.20	1	0.72
Expired Plates	368	17,151	50	0.13	3	0.93	361	19,562	60	0.13	2	0.90
Street Cleaning	282	14,820	50	0.11	4	1.05	271	15,766	60	0.11	3	0.97
Residential Permit Parking	222	12,858	60	0.09	5	0.96	179	13,033	75	0.09	5	0.97
Parking Prohibited Anytime	145	8,046	60	0.06	6	0.93	129	9,270	75	0.06	6	0.96
Rush Hour Parking	122	7,123	60	0.05	7	0.97	67	5,533	100	0.04	8	0.82
Rear And Front Plate Required	154	6,782	50	0.05	8	0.88	117	4,622	60	0.03	10	0.66
Park In Transit Stand	51	4,906	100	0.04	9	0.95	38	3,589	100	0.02	11	0.95
Within 15’ Of Fire Hydrant	46	4,549	100	0.03	10	0.99	43	5,119	150	0.03	9	0.80

Notes: This table reports average annual characteristics of the top ten revenue generating tickets from 2007-2011 and characteristics for the same set of tickets from 2012-2016. The type of ticket is listed in each row. Columns 1 and 7 report annual ticket volume, Columns 2 and 8 report annual ticket revenue, Columns 3 and 9 report the modal fine amount, Columns 4 and 10 report the revenue share of the listed ticket, along with the revenue share rank in Columns 5 and 11. Columns 6 and 12 report the ratio of revenue received and expected revenue, calculated as the base fine amount times ticket volume. The revenue share is less than one when collected revenue plus applicable late fees is less than the expected collected amount if all written tickets were paid on time, and is greater than one if collected revenue plus applicable late fees exceeds this expectation.

Table 2: Descriptive Statistics by Neighborhood: 2007-2011

	All Neighborhoods	Black Neighborhoods	Non-Black Neighborhoods
Panel A: Ticket-Level Outcomes	(1)	(2)	(3)
CPD Written	0.503	0.602	0.487
Sticker Ticket	0.076	0.149	0.064
Paid	0.527	0.471	0.548
Non-Payment Notice	0.228	0.308	0.197
Bankruptcy	0.021	0.039	0.014
Dismissed	0.224	0.182	0.240
CPD Written	0.572	0.655	0.541
Panel B: Neighborhood-Level Outcomes			
CPD Written	21,020	19,519	21,346
Sticker Ticket	3,176	4,842	2,814
Paid	1,674	2,280	1,542
Non-Payment Notice	723	1,492	556
Bankruptcy	66	188	40
Dismissed	713	882	676
CPD Written	1,816	3,173	1,521
Stickers Purchased	20,301	16,409	21,147
Panel C: Neighborhood Characteristics			
Share Black	0.305	0.934	0.168
Total Population	47,966	50,965	47,314
Total Vehicles	20,425	17,320	21,100

Notes: This table reports descriptive statistics by neighborhood type over the period 2007-2011. Panel A reports mean outcomes at the ticket-level and Panels B and C report mean annual outcomes at the neighborhood level. Column 1 reports overall means, Column 2 reports means in Black neighborhoods, defined as zip codes with a greater than seventy-five percent Black population share, and Column 3 reports means in non-Black neighborhoods. Sticker purchase data covers the period 2008-2011. Outcomes in Panel C are calculated using the 2007-2011 American Community Survey. We approximate total vehicles by aggregating bins based on survey responses and top code the highest bin as representing four vehicles.

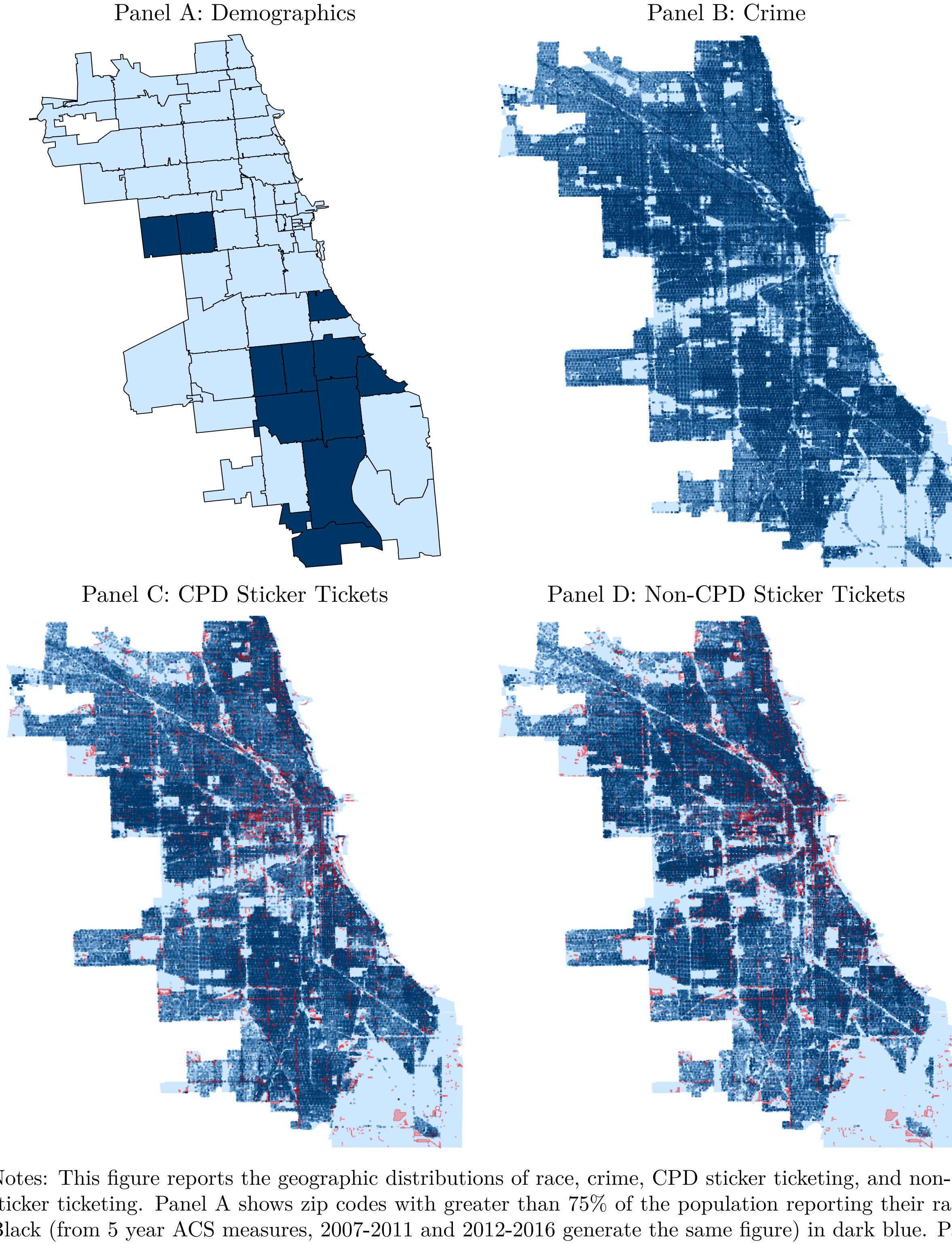
Table 3: Correlating Officer Policy Responses with Observable Characteristics

	δ_j Response			
Panel A: δ_j (Sticker, Black)	(1)	(2)	(3)	(4)
Male	-0.000			-0.000
	(0.003)			(0.003)
Age	-0.000**			-0.000
	(0.000)			(0.000)
Hispanic		-0.002		-0.002
		(0.003)		(0.003)
Asian or Native American		-0.001		-0.002
		(0.006)		(0.006)
Black		-0.009**		-0.008*
		(0.004)		(0.004)
Years Experience			-0.001***	-0.001
			(0.000)	(0.000)
Complaints per Year			-0.000	-0.001
			(0.002)	(0.002)
Tickets Issued per Year (00s)			0.002**	0.001*
			(0.001)	(0.001)
Panel B: δ_j (Sticker, Non-Black)				
Male	-0.000			-0.001
	(0.002)			(0.002)
Age	-0.000			-0.000
	(0.000)			(0.000)
Hispanic		0.002		0.002
		(0.002)		(0.002)
Asian or Native American		0.010		0.010*
		(0.006)		(0.006)
Black		0.001		0.002
		(0.002)		(0.002)
Years Experience			0.000	0.000
			(0.000)	(0.000)
Complaints per Year			0.001	0.001
			(0.002)	(0.002)
Tickets Issued per Year (00s)			0.001**	0.001**
			(0.001)	(0.001)
Observations	4,992	4,992	4,992	4,992
Unit Fixed Effects	Yes	Yes	Yes	Yes

Notes: This table reports regressions of officer-specific δ_j responses against officer-level observables. The sample includes only police officers in patrol units. The dependent variable in Panel A is the officer-specific δ_j for sticker tickets in Black neighborhoods and the dependent variable in Panel B is the corresponding δ_j for sticker tickets in Non-Black neighborhoods. Experience, complaints and tickets issued per year are all measured prior to the policy change. Robust standard errors are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

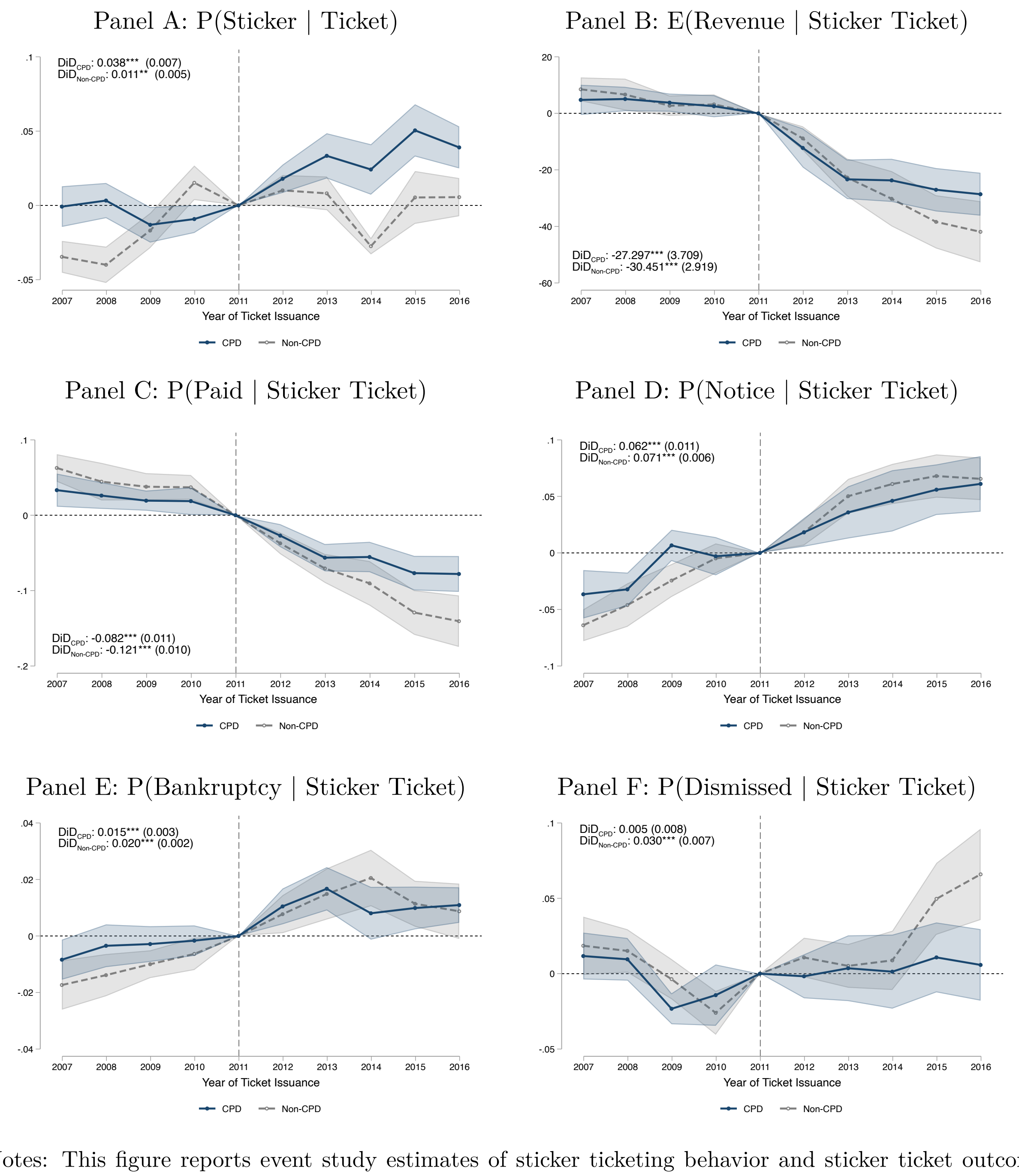
Appendix A: Additional Results

Appendix Figure A1: Maps of Sticker Tickets, Parking Amenities, Race, and Crime



Notes: This figure reports the geographic distributions of race, crime, CPD sticker ticketing, and non-CPD sticker ticketing. Panel A shows zip codes with greater than 75% of the population reporting their race as Black (from 5 year ACS measures, 2007-2011 and 2012-2016 generate the same figure) in dark blue. Panels B-D discretize Chicago into approximately 100,000 hexagons. Density is estimated with a quartic kernel over a bandwidth of 0.0005, with each color representing a quantile. Red polygons are parking amenities defined by Open Street Map. Panel B shows the distribution of reported criminal offenses with darker Blue areas reporting higher crime. Panels C and D show sticker ticket issuance by CPD and not CPD agents.

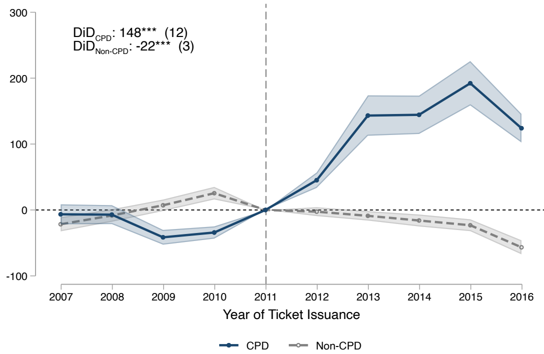
Appendix Figure A2: Event Study Estimates of Sticker Tickets and Outcomes at the Ticket-Level by Issuing Agency



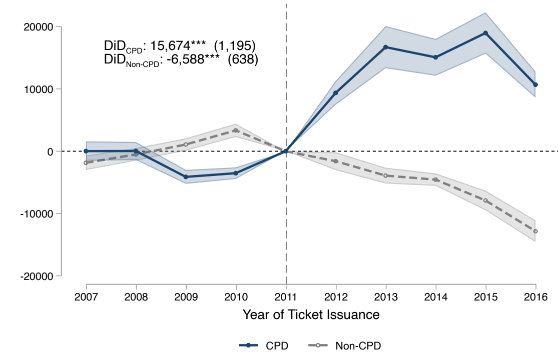
Notes: This figure reports event study estimates of sticker ticketing behavior and sticker ticket outcomes at the ticket level, estimated separately for the Chicago Police Department (CPD) and non-CPD agencies. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. The blue points report estimates for CPD and the gray points represent estimates for non-CPD. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level.

Appendix Figure A3: Event Study Estimates of Sticker Tickets and Outcomes at the Census Tract-Level by Issuing Agency

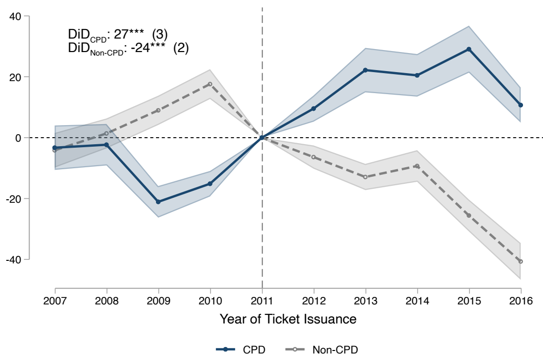
Panel A: Sticker Tickets



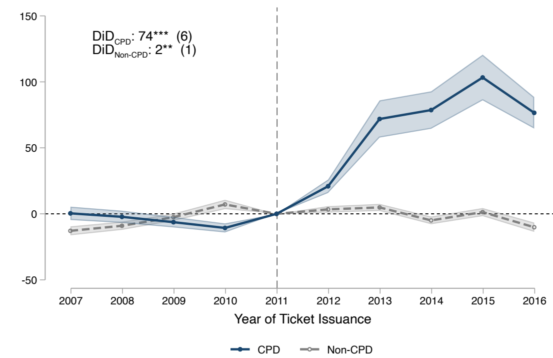
Panel B: Sticker Ticket Revenue



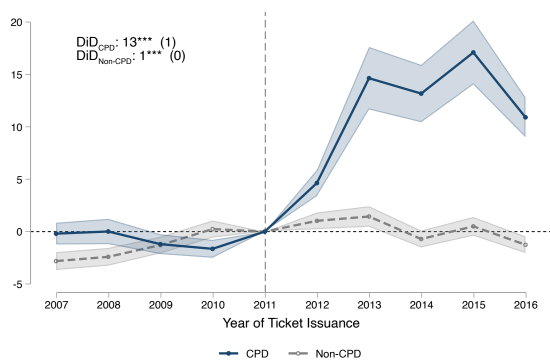
Panel C: Sticker Tickets Paid



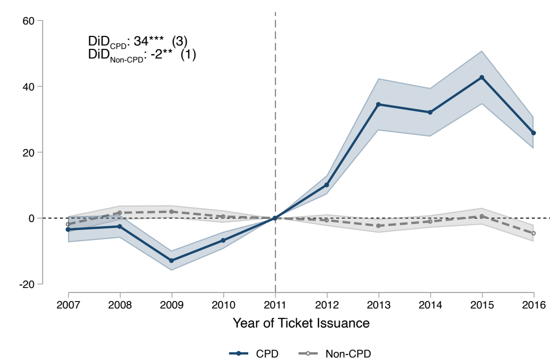
Panel D: Sticker Tickets Notice



Panel E: Sticker Tickets Bankrupt

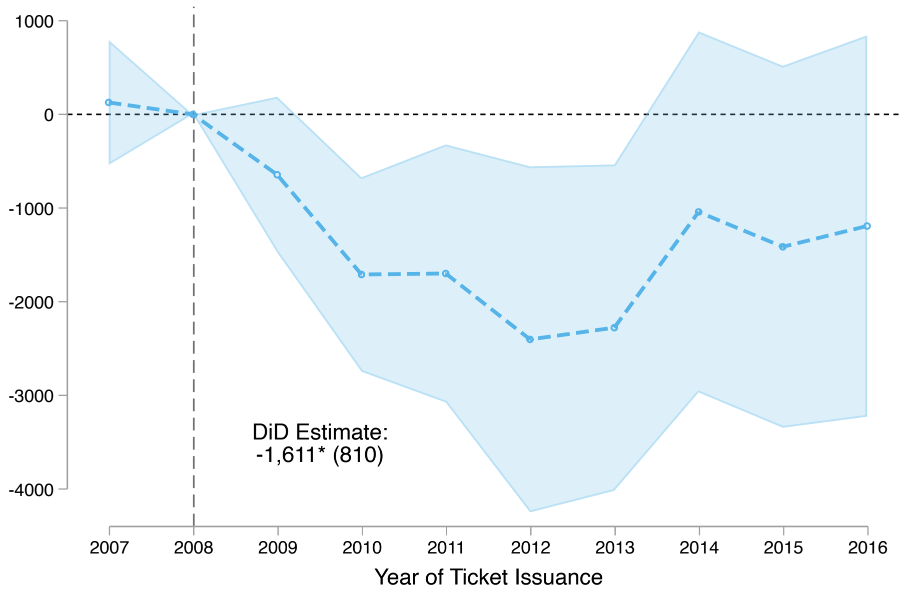


Panel F: Sticker Tickets Dismissed



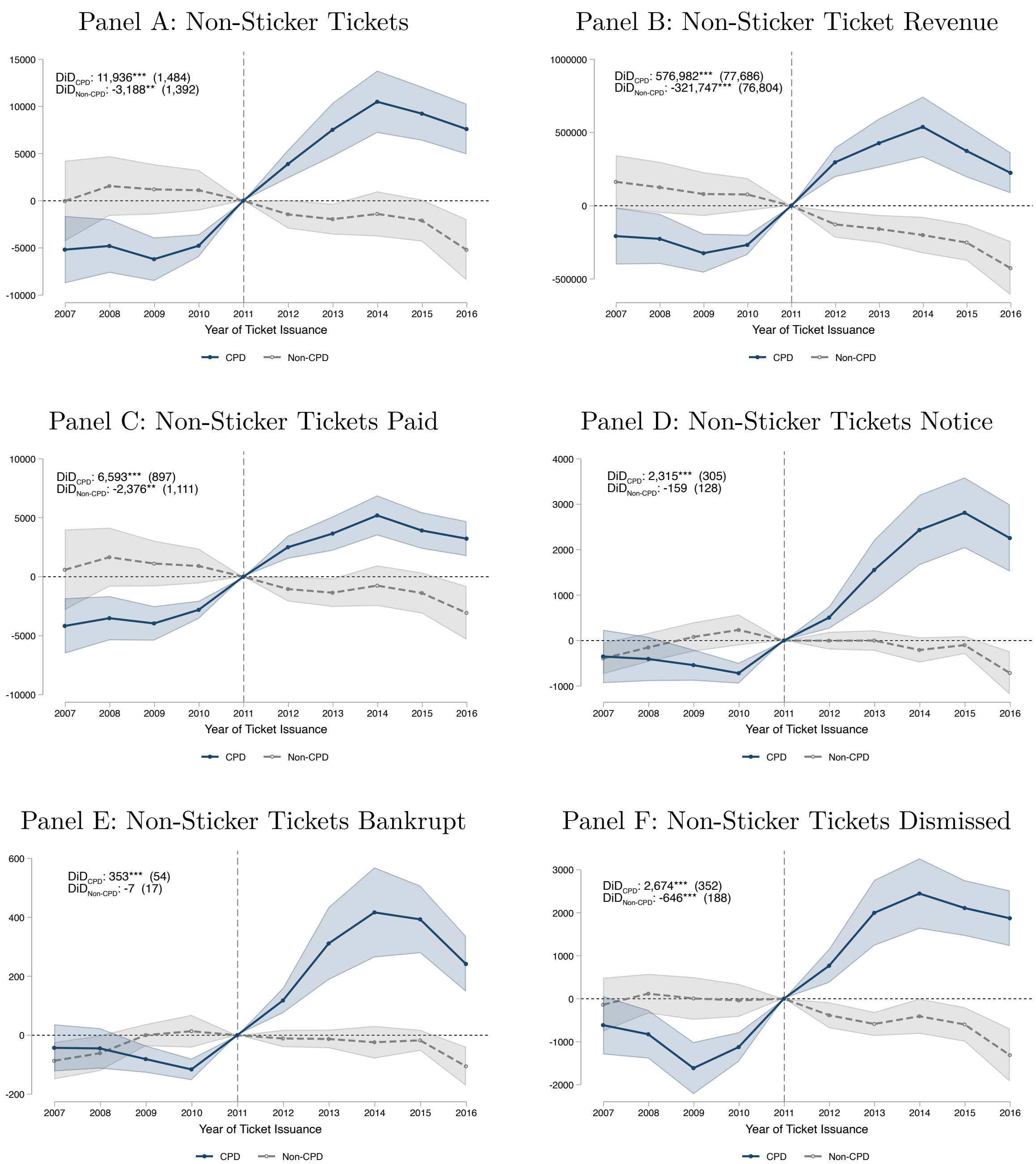
Notes: This figure reports event study estimates of sticker ticketing behavior and sticker ticket outcomes at the neighborhood (Census tract) level, estimated separately for the Chicago Police Department (CPD) and non-CPD agencies. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. The blue points report estimates for CPD and the gray points represent estimates for non-CPD. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level.

Appendix Figure A4: Event Study Estimates of Parking Meter Privatization on CPD vs Non-CPD Meter Ticketing



Notes: This figure reports event study estimates of parking meter tickets at the neighborhood level. Each point represents the effect of CPD parking meter ticketing activity compared to non-CPD activity, relative to 2008. The corresponding difference-in-differences estimate is displayed in the plot region. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

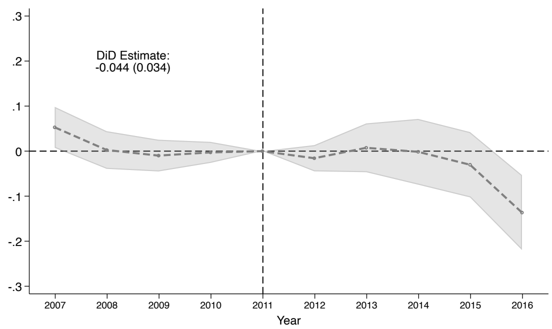
Appendix Figure A5: Event Study Estimates of Non-Sticker Tickets and Outcomes at the Neighborhood-Level by Issuing Agency



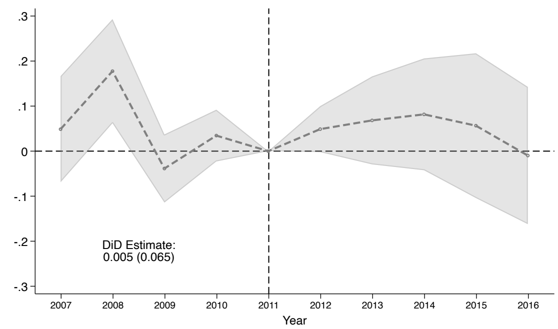
Notes: This figure reports event study estimates of non-sticker ticketing behavior and non-sticker ticket outcomes at the neighborhood level, estimated separately for the Chicago Police Department (CPD) and non-CPD agencies. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. The blue points report estimates for CPD and the gray points represent estimates for non-CPD. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level.

Appendix Figure A6: Event Study Estimates of Crimes Reported to Chicago Police Department and Arrests

Panel A: Ln(Crimes Reported)

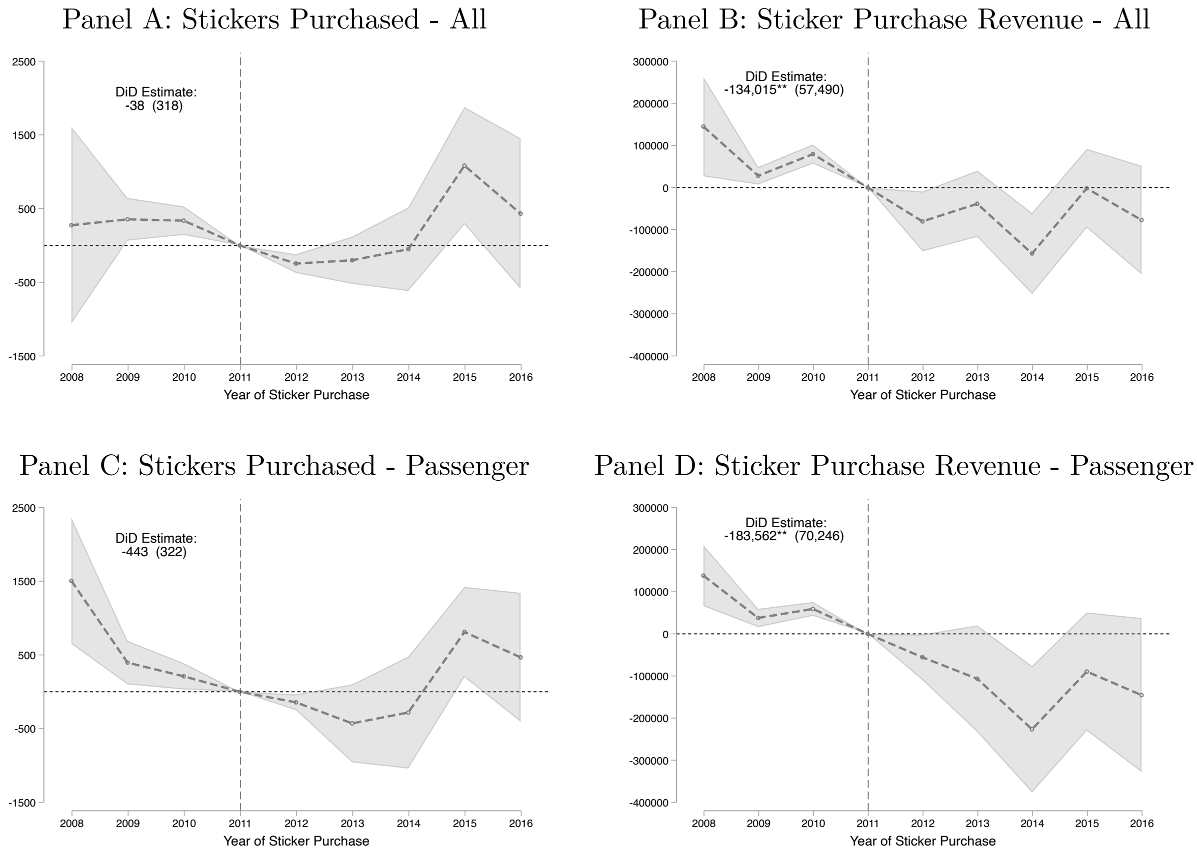


Panel B: Ln(Arrests)



Notes: This figure reports event study estimates of crimes reported to the Chicago Police Department and arrests made for those reported offenses. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. Panel A reports results for natural log of crimes reported to the Chicago Police Department and Panel B reports results for the natural log of arrests. Corresponding difference-in-differences estimates are reported in each panel. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

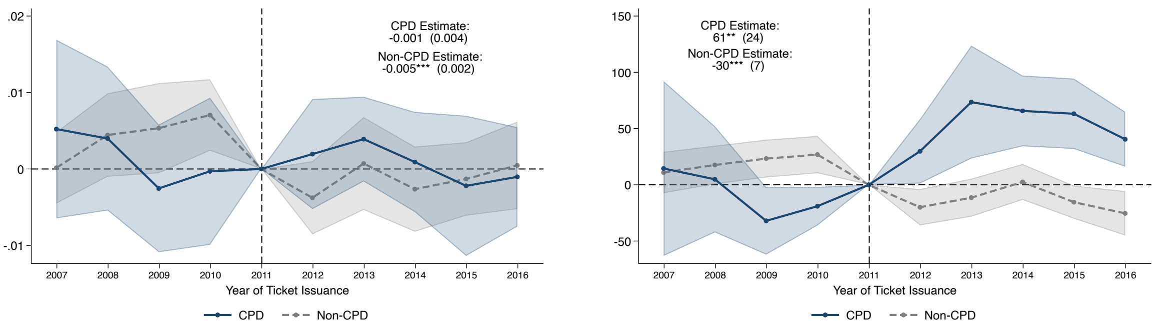
Appendix Figure A7: Event Study Estimates of Sticker Purchasing Behavior



Notes: This figure reports event study estimates of sticker purchases and sticker purchase revenue at the neighborhood level. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. All includes passenger, large vehicles, and motorcycles, the latter of which we exclude from the decomposition in the lower panels. Corresponding difference-in-differences estimates are reported in each panel. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Figure A8: Event Study Estimates of Sticker Tickets in Parking Amenities

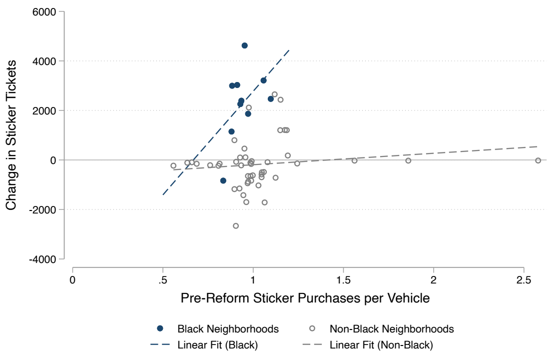
Panel A: P(Parking Amenity | Sticker) Panel B: Parking Amenity Sticker Tickets



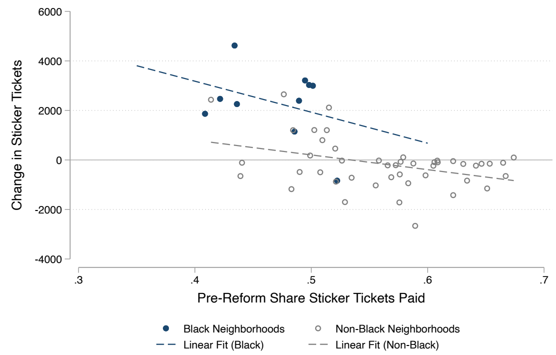
Notes: This figure reports event study estimates of sticker ticketing behavior in parking lots, estimated separately for the Chicago Police Department (CPD) and non-CPD agencies. 3.3% of CPD sticker tickets are issued in parking amenities (4.1% in non-Black zip codes, 2.2% in Black). 4.5% of not CPD sticker tickets are issued in parking amenities (5.3% in non-Black zip codes, 1.6% in Black). A ticket is deemed "in a parking amenity" if it occurs within 0.00013 lat/lon (about 15 meters) from a parking amenity or within a parking amenity as defined by Open Street Mapping. Results are robust to alternative buffer regions, even at a buffer of 0.0002, fewer than 9% of sticker tickets are in parking amenities. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. The blue points report estimates for CPD and the gray points represent estimates for non-CPD. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Figure A9: Additional Relationships Between Change in Sticker Ticketing and Neighborhood Characteristics

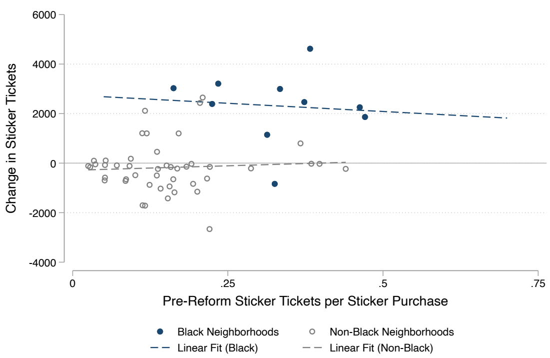
Panel A: Δ Tickets and Purchases per Vehicle - CPD



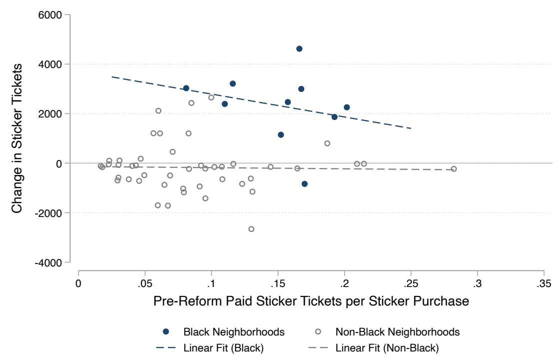
Panel B: Δ Tickets and Share Sticker Tickets Paid - CPD



Panel C: Δ Tickets and Tickets per Purchase - CPD

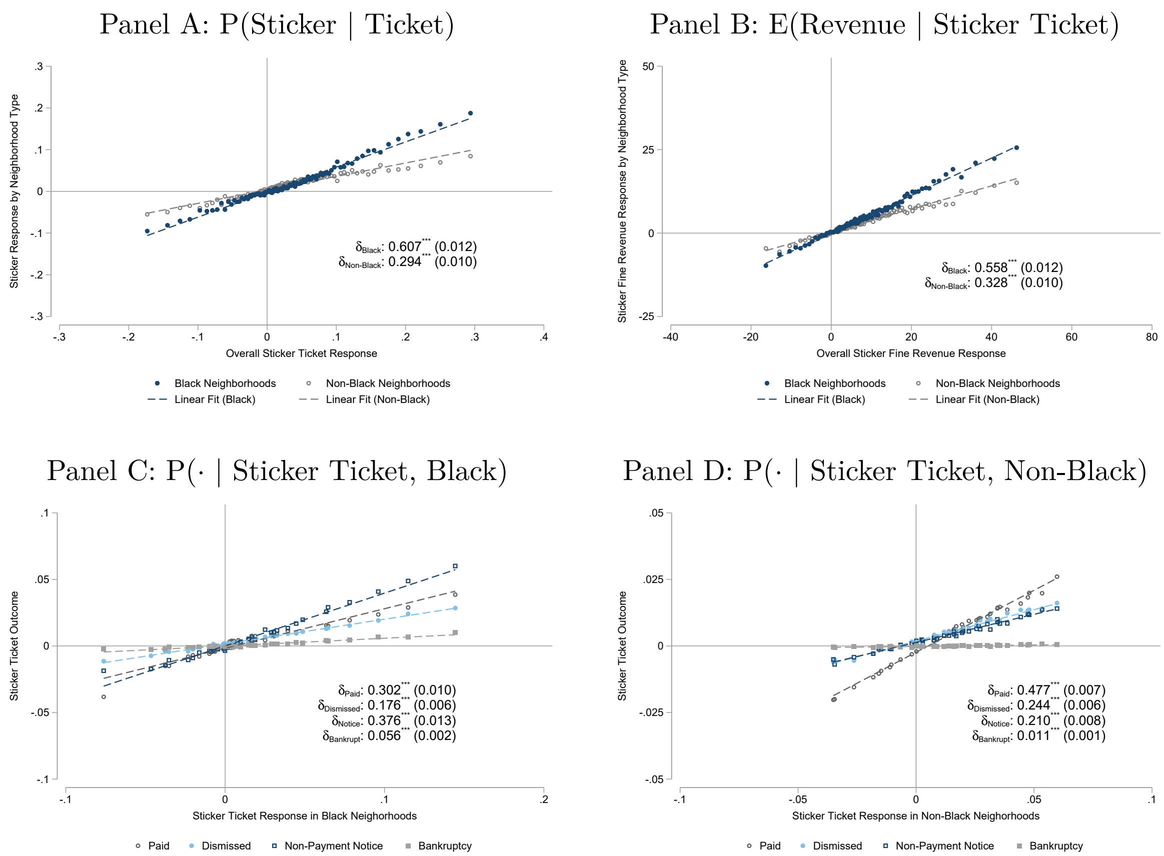


Panel D: Δ Tickets and Paid Tickets per Purchase - CPD



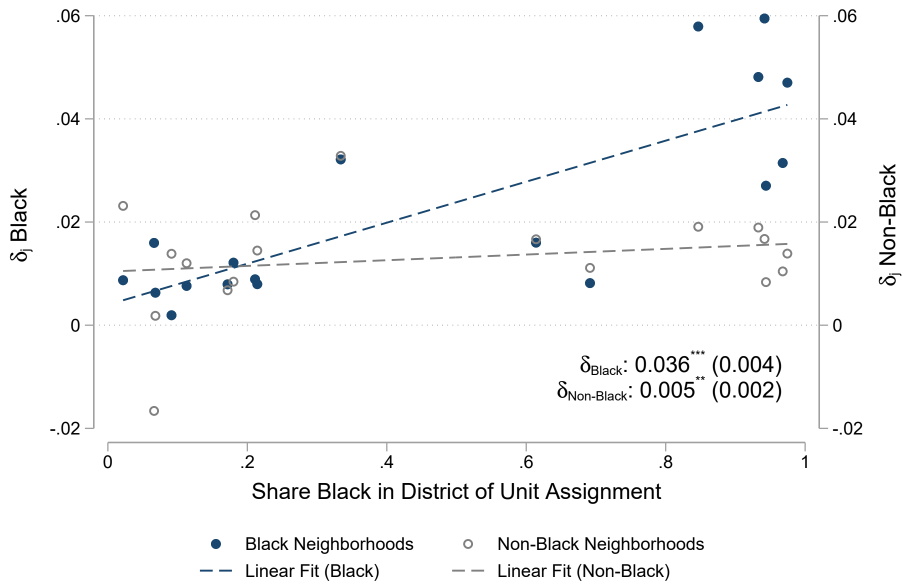
Notes: This figure reports joint distributions of neighborhood-level one-way difference estimates across different outcomes, along with neighborhood-level estimates with pre-reform characteristics. Each panel the change in sticker tickets issued by CPD against pre-reform neighborhood characteristics, calculated using data from 2008-2011. Each point represents a neighborhood, defined at the zip code level. Navy dots represent Black neighborhoods and gray circles represent non-Black neighborhoods. Dashed lines represent linear lines of best fit, estimated separately by neighborhood type.

Appendix Figure A10: Estimating and Decomposing Officer-Specific Responses to Sticker Fine Increase - Empirical Bayes-Adjusted



Notes: This figure plots Empirical Bayes-adjusted estimates of δ_j for different outcomes against each other. In Panels A and B we plot estimates of neighborhood race-specific δ_j responses against the overall race-agnostic δ_j response on the x-axis. In Panels C and D, we plot the race-specific sticker ticket outcomes against the race-specific sticker ticket responses, separately by race. There are 100 bins per outcome in the top two panels and 40 bins per outcome in the lower two panels. For exposition, we drop the first and last bin for each outcome. Reported coefficients estimated on the underlying officer-level estimates. Robust standard errors are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Figure A11: Officer-Specific Policy Responses and Assignment Demographics



Notes: This figure reports the correlation between officer-specific δ_j sticker ticket responses by neighborhood demographic group and the demographic composition of the modal unit assignment. Each point represents a separate unit assignment and plots the within-bin mean against share Black. Blue dots denote $\delta_j(Sticker, Black)$ responses (left axis) and gray circles represent $\delta_j(Sticker, Non-Black)$ responses (right axis). Dashed lines denote linear fits. Reported coefficients and standard errors are estimated on the underlying data. Robust standard errors are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Table A1: Annual Volume and Revenue from Top Ten Revenue Generating Tickets in Chicago by Neighborhood Demographic Composition

	2007-2011						2012-2016					
	Tickets (000s)	Revenue ($000s)	Avg. Fine	Rev. Share	Rev. Rank	Rev./ E[Rev.]	Tickets (000s)	Revenue ($000s)	Avg. Fine	Rev. Share	Rev. Rank	Rev./ E[Rev.]
Panel A: Black Neighborhoods	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
No City Sticker	48	5,034	120	0.24	1	0.87	71	8,593	200	0.33	1	0.60
Expired Plates	79	3,631	50	0.18	2	0.91	88	4,176	60	0.16	2	0.79
Street Cleaning	34	1,816	50	0.09	3	1.07	33	1,901	60	0.07	3	0.96
Rear And Front Plate Required	32	1,345	50	0.06	4	0.85	28	991	60	0.04	8	0.59
Park In Transit Stand	13	1,239	100	0.06	5	0.98	12	1,123	100	0.04	5	0.95
Residential Permit Parking	18	1,076	60	0.05	6	0.98	15	1,047	75	0.04	6	0.92
Parking Prohibited Anytime	15	913	60	0.04	7	0.99	20	1,419	75	0.05	4	0.96
Within 15' Of Fire Hydrant	8	800	100	0.04	8	0.98	9	1,034	150	0.04	7	0.76
Rush Hour Parking	14	796	60	0.04	9	0.97	9	662	100	0.02	10	0.77
Expired Meter	11	566	50	0.03	10	1.02	8	426	50	0.02	14	1.02
Panel B: Non-Black Neighborhoods												
Expired Meter	393	18,966	50	0.16	1	0.97	274	14,839	50	0.12	3	1.08
No City Sticker	129	13,580	120	0.12	2	0.88	134	20,982	200	0.17	1	0.78
Expired Plates	288	13,520	50	0.12	3	0.94	274	15,386	60	0.13	2	0.94
Street Cleaning	248	13,004	50	0.11	4	1.05	238	13,865	60	0.12	4	0.97
Residential Permit Parking	204	11,782	60	0.10	5	0.96	164	11,986	75	0.10	5	0.97
Parking Prohibited Anytime	129	7,133	60	0.06	6	0.92	109	7,851	75	0.07	6	0.96
Rush Hour Parking	108	6,327	60	0.05	7	0.98	59	4,871	100	0.04	8	0.83
Rear And Front Plate Required	122	5,437	50	0.05	8	0.89	89	3,631	60	0.03	10	0.68
Within 15' Of Fire Hydrant	38	3,749	100	0.03	9	0.99	34	4,085	150	0.03	9	0.81
Park In Transit Stand	39	3,667	100	0.03	10	0.94	26	2,465	100	0.02	11	0.95

Notes: This table reports average annual characteristics of the top ten revenue generating tickets from 2007-2011 and characteristics for the same set of tickets from 2012-2016, separately by neighborhood demographic composition. Panel A reports statistics for Black neighborhoods, define as neighborhoods whose population is at least seventy-five percent Black, and Panel B reports statistics for Non-Black neighborhoods. The type of ticket is listed in each row. Columns 1 and 7 report annual ticket volume, Columns 2 and 8 report annual ticket revenue, Columns 3 and 9 report the modal fine amount, Columns 4 and 10 report the revenue share of the listed ticket, along with the revenue share rank in Columns 5 and 11. Columns 6 and 12 report the ratio of revenue received and expected revenue, calculated as the base fine amount times ticket volume. The revenue share is less than one when collected revenue plus applicable late fees is less than the expected collected amount if all written tickets were paid on time, and is greater than one if collected revenue plus applicable late fees exceeds this expectation.

Appendix Table A2: Robustness Checks of Neighborhood-Level Regressions

Panel A: CPD	(1)	(2)	(3)	(4)	(5)
Tickets	2,490***	2,490***	2,308***	2,166***	2,347***
	(461)	(464)	(414)	(385)	(508)
Revenue	271,341***	271,341***	256,656***	249,193***	260,740***
	(50,779)	(51,102)	(47,257)	(42,036)	(56,236)
Paid	363***	363***	283***	355***	336***
	(113)	(114)	(98)	(100)	(122)
Notice	1,316***	1,316***	1,279***	1,088***	1,249***
	(220)	(222)	(198)	(189)	(244)
Bankruptcy	236***	236***	222***	195***	222***
	(38)	(38)	(33)	(33)	(42)
Dismissed	574***	574***	522***	528***	539***
	(107)	(108)	(102)	(87)	(117)
Panel B: Non-CPD					
Tickets	-294***	-294***	-392***	-323***	-314***
	(82)	(83)	(133)	(78)	(85)
Revenue	-77,131***	-77,131***	-83,071***	-80,922***	-80,994***
	(17,194)	(17,303)	(17,613)	(17,588)	(17,189)
Paid	-374***	-374***	-358***	-371***	-383***
	(64)	(65)	(95)	(58)	(68)
Notice	58*	58*	-23	40	53
	(34)	(34)	(32)	(26)	(37)
Bankruptcy	29***	29***	3	23***	28***
	(7)	(7)	(11)	(6)	(8)
Dismissed	-9	-9	-16	-14	-12
	(20)	(20)	(15)	(18)	(21)
Zip Fixed Effects	No	Yes	Yes	Yes	Yes
Year Fixed Effects	No	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	C-S	OLS	OLS
Black Majority Cutoff	0.75	0.75	0.75	0.50	0.90

Notes: This table reports robustness checks of our neighborhood difference-in-differences (DiD) estimates. Panel A reports results for CPD-written tickets and Panel B reports results for non-CPD-written tickets. The outcome is listed in each row. Each cell corresponds to a different point estimate. Column 1 estimates the simple DiD without any additional fixed effects, Column 2 reproduces our main text estimate, Column 3 uses the estimator from Callaway and Sant'Anna (2021), Column 4 changes the cutoff to define Black neighborhoods to 50%, and Column 5 changes the same threshold to 90%. Standard errors clustered at the zip code level are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Table A3: Decomposing Differential Outcomes by Owner Zip Code Demographics

	Tickets	Revenue	Paid	Notice	Bankruptcy	Dismissed
Panel A: CPD	(1)	(2)	(3)	(4)	(5)	(6)
Main:	2,490***	271,341***	363***	1,316***	236***	574***
	(464)	(51,102)	(114)	(222)	(38)	(108)
Black Zip Owner:	1,464***	196,497***	53	968***	184***	257***
	(278)	(32,861)	(66)	(151)	(25)	(43)
Non-Black Zip Owner:	772***	65,194***	269***	323***	48***	132***
	(179)	(21,376)	(57)	(85)	(15)	(36)
Panel B: Non-CPD						
Main:	-294***	-77,131***	-374***	58*	29***	-9
	(83)	(17,303)	(65)	(34)	(7)	(20)
Black Zip Owner:	-83*	25,558***	-183***	75***	24***	1
	(46)	(7,683)	(41)	(24)	(5)	(7)
Non-Black Zip Owner:	-212***	-96,605***	-175***	-15	5*	-27***
	(46)	(12,785)	(31)	(13)	(3)	(10)

Notes: This table decomposes our main difference-in-differences estimates into outcomes experienced by owners in majority (>75 percent) Black neighborhoods and those in non-Black majority neighborhoods. We interact each outcome in the column title with indicators for $Black_i$ and $(1 - Black_i)$. The "Main" row reproduces our main text estimate and the corresponding "Black" and "Non-Black" rows decompose the Main outcome following the previous description. Due to missing owner information for some tickets, the decomposition will not exactly add to the full sample estimate. Panels A and C report results for CPD-written tickets and Panels B and D report results for non-CPD-written tickets. The upper panels report ticket-level estimates and the lower panels report neighborhood-level estimates. Standard errors clustered at the zip code level are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Table A4: Testing Departmental Responses to Alternative Treatment Margins

	Sticker Tickets					
Panel A: CPD	(1)	(2)	(3)	(4)	(5)	(6)
Black × Post	2,490***					1,129
	(464)					(720)
High Income × Post		-689**				-328
		(268)				(281)
High Crime × Post			2,005***			1,206**
			(426)			(523)
High Sticker Ticket Rate × Post				1,755***		691*
				(460)		(369)
High Sticker Ticket Payment Rate × Post					-863***	30
					(275)	(268)
Panel B: Non-CPD						
Black × Post	-294***					-40
	(83)					(211)
High Income × Post		80				139
		(117)				(125)
High Crime × Post			-256***			-229
			(76)			(141)
High Sticker Ticket Rate × Post				-260***		-175
				(75)		(115)
High Sticker Ticket Payment Rate × Post					-11	-189*
					(106)	(101)

Notes: This table presents difference-in-differences estimates using alternative treatment definitions. Panel A reports results for CPD-written tickets, and Panel B reports results for non-CPD-written tickets. The corresponding interaction is listed in each row. The outcome in all columns is the number of sticker tickets. Non-Black alternative treatment definitions are defined as being in the upper quartile or not. All regressions include zip code and year fixed effects. Standard errors clustered at the zip code level are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Table A5: Decomposing Differential Outcomes by Non-Race Neighborhood Characteristics Within Black Neighborhoods

	Income		Crime Rate		Sticker Ticket Rate		Sticker Ticket Payment Rate	
	High	Low	High	Low	High	Low	High	Low
Panel A: CPD	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Sticker	3,055***	1,925***	3,017***	1,964***	2,592***	1,323***	1,206***	2,516***
	(537)	(630)	(463)	(699)	(476)	(276)	(284)	(450)
p-value	0.164		0.200		0.000		0.000	
Panel B: Non-CPD								
Sticker	-338***	-251**	-328***	-261**	-373***	-242***	-4	-397***
	(106)	(99)	(75)	(126)	(137)	(70)	(87)	(100)
p-value	0.489		0.600		0.336		0.005	

Notes: This table reports difference-in-differences results which decompose the differential response in Black neighborhoods along other non-race characteristics, estimated at the neighborhood-level. Each non-race characteristic is defined in the pre-reform period and splits the subsample of Black neighborhoods into above- and below-median groups based on the statistic listed in the column title. Sticker ticket rate is the fraction of neighborhood tickets which are sticker tickets. Sticker ticket payment rate is the fraction of neighborhood sticker tickets which are paid. Panel A reports results for CPD-written tickets and Panel B reports results for non-CPD-written tickets. Listed p-values test for differences between coefficient estimates in Black neighborhoods. All regressions include zip code and year fixed effects. Standard errors clustered at the zip code level are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix Table A6: Decomposing Relative Contributions of Differential Compliance and Enforcement in Sticker Ticketing Gap

	Black Neighborhoods	Non-Black Neighborhoods	Gap
	(1)	(2)	(1)-(2)
Share Vehicles Unregistered	0.196	0.200	-0.004
Δ Enforcement Share	0.689	-0.047	0.737
Δ Compliance Share	0.123	0.121	0.002
Total Vehicles	17,320	21,893	-4,574
Unregistered Vehicles	3,356	3,725	-369

Notes: This table calculates statistics describing the size of the unregistered vehicle stock in the pre-reform period, separately by neighborhood type. The first row calculates the share of vehicles without a valid sticker as of the end of the grace period. The number of unregistered vehicles is estimated using the 2007-2011 American Community Survey. We drop a small handful of neighborhoods with negative estimated non-compliance rates due to measurement error in number of vehicles. The second and third rows estimate the share of unregistered vehicles who were affected by the reform, dividing one-way differences in CPD enforcement and consumer purchases by the number of unregistered vehicles. Column 1 reports these estimates for Black neighborhoods, Column 2 for non-Black neighborhoods, and Column 3 reports the difference between Columns 1 and 2.

Appendix Table A7: Correlating Officer Policy Responses with Observable Characteristics - All Officers

	δ_j Response			
Panel A: δ_j (Sticker, Black)	(1)	(2)	(3)	(4)
Male	0.001			0.001
	(0.003)			(0.003)
Age	-0.000**			-0.000
	(0.000)			(0.000)
Hispanic		-0.001		-0.001
		(0.003)		(0.003)
Asian or Native American		-0.002		-0.003
		(0.006)		(0.006)
Black		-0.008**		-0.007*
		(0.004)		(0.004)
Years Experience			-0.000**	-0.000
			(0.000)	(0.000)
Complaints per Year			0.000	-0.000
			(0.002)	(0.002)
Tickets Issued per Year (00s)			0.001*	0.001
			(0.001)	(0.001)
Panel B: δ_j (Sticker, Non-Black)				
Male	-0.000			-0.001
	(0.002)			(0.002)
Age	-0.000			-0.000
	(0.000)			(0.000)
Hispanic		0.001		0.001
		(0.002)		(0.002)
Asian or Native American		0.005		0.005
		(0.005)		(0.005)
Black		0.000		0.001
		(0.002)		(0.002)
Years Experience			0.000	0.000
			(0.000)	(0.000)
Complaints per Year			0.001	0.001
			(0.001)	(0.001)
Tickets Issued per Year (00s)			0.001	0.001
			(0.001)	(0.001)
Observations	6,153	6,153	6,153	6,153
Unit Fixed Effects	Yes	Yes	Yes	Yes

Notes: This table reports regressions of officer-specific δ_j responses against officer-level observables. The sample includes all police officers. The dependent variable in Panel A is the officer-specific δ_j for sticker tickets in Black neighborhoods and the dependent variable in Panel B is the corresponding δ_j for sticker tickets in Non-Black neighborhoods. Experience, complaints and tickets issued per year are all measured prior to the policy change. Robust standard errors are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Appendix B: Model of Officer Behavior

In this section, we present a simple model of CPD and non-CPD officer behavior to provide a simple microfoundation for understanding the differences between officers across departments and illustrating how the reform impacts various channels which influence officer behavior.

Non-CPD Behavior: We operationalize non-CPD agencies as maximizing ticket volume:

$$\max_{s_n} \; s_n v_n - \underbrace{c(s_n)}_{\substack{\text{Search}\\\text{cost}}}$$

where s_n is a search in neighborhood n, v_n is the violation rate in neighborhood n, and $c(s_n) = c_n s_n^2/2$ is the neighborhood-specific convex search cost. Standard first order conditions equate the violation rate with the marginal cost of search. Notably, this solution is independent of the expected value of the ticket, including both the fine amount and payment probability.

CPD Behavior: In contrast, CPD maximizes revenue and other objectives:

$$\max_{s_n} \; s_n(\underbrace{f * p_n * v_n}_{\text{E[Revenue]}}) - \underbrace{c(s_n)}_{\substack{\text{Search}\\\text{cost}}}$$

In this setup E[Revenue]= $(f * p_n * v_n)$, where f is the fine, p_n is the probability a ticket gets paid and $c(s_n) = c_n s_n^2/2$ is the cost of search, including the opportunity cost.

The first order conditions of this problem should equate marginal expected revenue to marginal costs. Thus, there will more tickets in neighborhood n if, all else equal there are higher violation rates, higher payment rates, or lower marginal costs. The fine increase will affect both the level and distribution of tickets. Writing out the FOCs:

$$\underbrace{f * p_n * v_n}_{\substack{\text{Marginal}\\\text{revenue}}} = \underbrace{c_n s_n^*}_{\substack{\text{Marginal}\\\text{cost}}}$$

We are interested in the change in search with respect to a change in fine, $\frac{\delta s_n^*}{\delta f}$, implying:

$$\underbrace{p_n * v_n}_{\substack{\text{Revenue}\\\text{effect}}} + \underbrace{f * \frac{\delta p_n}{\delta f} * v_n}_{\substack{\text{Payment rate}\\\text{effect}}} + \underbrace{f * p_n * \frac{\delta v_n}{\delta f}}_{\substack{\text{Deterrence}\\\text{effect}}} = c_n \frac{\delta s_n^*}{\delta f}$$

The first term of this equation $(p_n * v_n)$ can be thought of as a revenue effect, which we find as smaller in Black neighborhoods. The second term, the payment rate effect, is likely smaller in Black neighborhoods because of a larger reduction in repayment, with low violation rates across all neighborhoods. The final term, we can label a deterrence effect, which we find is relatively small. We conclude that the larger increase in tickets in Black neighborhoods is likely due to smaller marginal costs for CPD officers.